Injective analytic mappings of low order

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Abstract

Let $\boldsymbol{n} : (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0)$ be a holomorphic map germ admitting an injective representative. We prove that $(d\boldsymbol{n})_0 \neq 0$, as conjectured by Lê. Building on the author's earlier reduction theorem [PC26], this work completes the proof of the conjecture by excluding maps of orders 2, 3, and 4. The proofs combine successive analytic changes of source and target coordinates, Weierstrass preparation, and the analysis of weighted initial forms with techniques from singularity theory, including semicontinuity of the spectrum. In dialogue with the author, ChatGPT developed most of the mathematical ideas, repaired a gap in the author's initial argument for order 4, and wrote most of the manuscript subsequently refined by the author.

1 Introduction

Let

$$\boldsymbol{n} : (\mathbb{C}^2, 0) \longrightarrow (\mathbb{C}^3, 0), \qquad (x, y) \longmapsto (n_1(x, y), n_2(x, y), n_3(x, y))$$

be a holomorphic map germ admitting an injective representative. Its *order* is $m_{\boldsymbol{n}} = \operatorname{ord}_0(\boldsymbol{n}) = \min_{1\leq j\leq 3} \operatorname{ord}_0(n_j)$. The reduction theorem [PC26, Theorem 1.1] states that $(d\boldsymbol{n})_0 = 0$ implies $m_{\boldsymbol{n}} \in \{2, 3, 4\}$. The purpose of this paper is to exclude these three orders and thereby prove Lê's conjecture. We refer to the introduction of [PC26] for the geometric formulation in terms of links of reduced surface germs, the equivalence of the two formulations, and the history of the problem.

Theorem 1.1. *Every holomorphic map germ* $\boldsymbol{n} : (\mathbb{C}^2, 0) \longrightarrow (\mathbb{C}^3, 0)$ *that admits an injective representative satisfies* $(d\boldsymbol{n})_0 \neq 0$.

Strategy and organization of the proof

Suppose that $(d\boldsymbol{n})_0 = 0$. Choose linear coordinates on the target as in Section 2, and write $\boldsymbol{n} = (v, u, w)$ in these coordinates, so that (v, u) is finite. Put $X = \boldsymbol{n}(\mathbb{C}^2)$, $V = \{v = 0\}$, $m_{\boldsymbol{n}} \in \{2, 3, 4\}$, and $m_X = \operatorname{mult}_0(X)$. The geometric question is whether the third coordinate can distinguish all the points with the same first two coordinates.

To see this on a curve, choose $\delta > 0$ sufficiently small and then $\eta > 0$, and put $\overline{D}_\delta = \{z \in \mathbb{C} : |z| \leq \delta\}$. For $0 < |s| < \eta$, choose a Milnor representative V_s of $\{v = s\}$ so that

$$u : V_s \longrightarrow \overline{D}_\delta, \qquad p \longmapsto u(p)$$

is a proper branched covering of degree m_X, unramified on the boundary, as in [PC26, proof of Proposition 2.16]. This is the covering used to obtain the upper bound on transverse multiplicity in the reduction theorem. Write $W_s = \boldsymbol{n}(V_s)$ for the corresponding representative of the target section. On V_s, the map $\boldsymbol{n}$ is

$$\boldsymbol{n}|_{V_s} : V_s \longrightarrow W_s, \qquad p \longmapsto (s, u(p), w(p)).$$

For $p, q \in V_s$, the first coordinates agree. Consequently,

$$\boldsymbol{n}(p) = \boldsymbol{n}(q) \quad \Longleftrightarrow \quad u(p) = u(q) \text{ and } w(p) = w(q). \tag{1.2}$$

Thus injectivity requires w to take distinct values at every pair of distinct points in a fiber of $u|_{V_s}$. We call a pair of distinct points with the same image under a map a *collision* of that map. The same requirement applies to every fiber of the finite map (v, u).

The proof develops this separation requirement in three steps.

1. *Count the ramification and prepare a critical component.* Writing $\mu(V, 0)$ for the Milnor number of V, the Riemann–Hurwitz formula gives total ramification $m_X + \mu(V, 0) - 1$ for $u|_{V_s}$. Choose a component Γ_i of the critical locus whose image has maximal transverse multiplicity m_{tr}, and put $a_i = i_0(v, \Gamma_i)$, as in Section 2. The numerical inequalities and base-point formulas yield Proposition 3.6: Γ_i is smooth and $a_i = m_{\boldsymbol{n}}$. The component Γ_i meets V_s in a_i points. At each of them the image has a transverse branch of multiplicity m_{tr}, so the local degree of u is at least m_{tr}. These points contribute at least $a_i(m_{\mathrm{tr}} - 1)$ to the total ramification, giving

   $$a_i(m_{\mathrm{tr}} - 1) \leq m_X + \mu(V, 0) - 1.$$

   The same transverse multiplicity forces the Jacobian minors of $\boldsymbol{n}$ to vanish to order at least $m_{\mathrm{tr}} - 1$ along Γ_i.

2. *Determine the first separation of the roots.* Make Γ_i the axis $y = 0$ and write V as the zero set of a Weierstrass polynomial of degree $m_{\boldsymbol{n}}$ in x, with the roots translated so that their mean is zero. Rescale at the first order at which its roots separate as $y \to 0$. The resulting weighted initial forms must satisfy both the numerical inequalities and the divisibility of the Jacobian minors from Step 1. These restrictions exclude order four in Section 4. In order three they leave two possible pairs of initial forms. Order two requires a further analytic preparation. Here $m_{\boldsymbol{n}}$ counts the roots of the Weierstrass equation. The covering $u|_{V_s}$ has degree m_X, which can be much larger.

3. *Show that the third coordinate fails to separate a pair.* For the remaining orders, study pairs of points with the same first two coordinates and the difference of their third coordinates. Subtracting a function of the first two coordinates does not change this difference. When the first two initial forms define a finite map, Lemma 5.16 uses such subtractions to expose a third initial form outside their polynomial algebra. The fiber products of the pairs of initial forms describe the pairs of points that the third initial form must separate. In order three, symmetries and a derivative identity force equal third-coordinate values at distinct points. In the other projective curve arguments, the difference is a section whose zeros are counted with multiplicity, and the bounds prevent all its zeros from lying on the diagonal. An isolated zero corresponding to distinct points lifts, by Lemma 3.10, to collisions of the analytic map arbitrarily close to the origin. Order two also uses a local fiber product disjoint from the diagonal. Injectivity would confine the zeros of the third-coordinate difference to its limiting fiber. A second rescaling shows that this constraint forces a collision. Each collision violates (1.2).

Section 3 establishes the common preparation, the weighted estimates, and the lifting criterion used in these steps. The three subsequent sections carry out these steps in orders four, three, and two, respectively.

AI disclosure

This paper was essentially written by ChatGPT, through several conversations and extended, largely autonomous work in dialogue with the author. ChatGPT developed the proofs for $m_{\boldsymbol{n}} = 2, 3, 4$, including the repair of a gap in the author's initial argument for $m_{\boldsymbol{n}} = 4$. The author's contribution to this paper has consisted primarily of refining and substantially shortening the generated text, organizing the ideas, and reviewing and verifying the arguments. The author takes responsibility for all the arguments in the final manuscript.

The proofs below rely, sometimes, on the results of the companion manuscript *A reduction theorem for Lê's conjecture* [PC26] which was part of the prompt that ChatGPT used to solve these cases. In the course of this work, we have not yet found a proof of the cases $m_{\boldsymbol{n}} = 2, 3, 4$ that does not rely on that work.

Although the proofs of the three cases are presented in a common framework, some auxiliary results remain specific to individual cases. Examples include Propositions 5.8 and 6.27. The proof of Proposition 6.27 is long because its steps depend closely on the particular form of the map. This dependence makes it difficult to divide the proof into shorter arguments that can be reused elsewhere in the paper. There are others like this.

We draw special attention to Lemma 6.6, whose proof uses semicontinuity of the spectrum and whose statement will trap a specialist in plane curve singularities an afternoon or two. We have not found an elementary proof of this lemma.

2 Input from the reduction theorem

We use the notation of the reduction theorem. Throughout the proof, let $\boldsymbol{n} = (n_1, n_2, n_3) : (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0)$ be a holomorphic map germ admitting an injective representative and satisfying $(d\boldsymbol{n})_0 = 0$. Write $(X, 0)$ for its reduced image germ. We fix an injective representative of $\boldsymbol{n}$ and a compatible representative of X, denoted by the same symbols. Choose coordinates (x, y) on $(\mathbb{C}^2, 0)$ and put

$$B = \mathbb{C}\{x, y\}, \qquad \mathfrak{m} = (x, y)B, \qquad I = (n_1, n_2, n_3)B, \qquad m_{\boldsymbol{n}} = \operatorname{ord}_0 I.$$

The map $\boldsymbol{n} : (\mathbb{C}^2, 0) \to (X, 0)$ is the normalization map, and I is $\mathfrak{m}$-primary. Starting with I, at each blowup we pull back local generators of the current ideal and divide them by their highest common power of a local equation of the new exceptional curve. The ideal generated by these quotients is the new *weak transform*. We blow up the points where this ideal is not the unit ideal, and stop when it is the unit ideal. This is the *principalization* used in the reduction theorem. Let $\mathcal{C}$ be the set of centres of these blowups, called the *cluster of base points*, and let $m_{\boldsymbol{n},p}$ be the order of the weak transform at p, as in [PC26, Notation 2.4]. For $f, g \in B$, write $J(f, g) = f_x g_y - f_y g_x$ for their *Jacobian determinant* with respect to (x, y). For other coordinates (s, t), we write $J_{s,t}(f, g) = f_s g_t - f_t g_s$. The subscript records the differentiation variables in order, so $J_{t,s}(f, g) = -J_{s,t}(f, g)$. For $f, g \in B$ whose zero sets have no common curve component, their *intersection multiplicity* at the origin is

$$i_0(f, g) = \dim_{\mathbb{C}} B/(f, g)B.$$

For $f \in B$ and a reduced plane curve germ $\Gamma \subset (\mathbb{C}^2, 0)$ having no component in $\{f = 0\}$, choose a reduced equation $h = 0$ for Γ and set $i_0(f, \Gamma) = i_0(f, h)$.

Choose general linear forms λ_1, λ_2 on the target as in [PC26, Sections 2.1–2.2], and complete them to target coordinates with a third linear form λ_3. Put $v = \lambda_1 \circ \boldsymbol{n}$, $u = \lambda_2 \circ \boldsymbol{n}$, and $w = \lambda_3 \circ \boldsymbol{n}$. Thus $V = \{v = 0\}$ is reduced, its strict transform has multiplicity $m_{\boldsymbol{n},p}$ at each $p \in \mathcal{C}$, and its final

strict transform is smooth, transverse to the exceptional divisor, and disjoint from its crossings. There are no infinitely near singular points of V outside $\mathcal{C}$. Moreover, $\{v = u = 0\} = \{0\}$, and u has minimal order on each branch of V among the elements of I. Write $W = X \cap \{\lambda_1 = 0\}$, and denote the *delta invariant* and *Milnor number* of a reduced plane curve germ by δ and μ, respectively, following [GLS07, p. 206 and Definition I.2.1]. Let b be the number of branches of V, let $m_X = \mathrm{mult}_0(X) = \mathrm{mult}_0(W)$, and put $M_{\boldsymbol{n}} = m_X - 2\delta(V, 0)$.

Write $\Sigma = \mathrm{Sing}(X) = \bigcup_{i=1}^{c} \Sigma_i$ for the decomposition into irreducible components, and put $\Gamma_i = (\boldsymbol{n}^{-1}(\Sigma_i))_{\mathrm{red}}$. The curves Γ_i are the irreducible components of the critical locus of $\boldsymbol{n}$. As in the reduction theorem, the *transverse multiplicity* m_i along Σ_i is the multiplicity of a transverse plane branch of X at a general point of Σ_i. Set

$$a_i = i_0(v, \Gamma_i), \qquad m_{\mathrm{tr}} = \max_i m_i \geq 2.$$

We also choose v so that the final strict transform of V avoids the strict transforms of the finitely many Γ_i over the origin. The strict transforms of the Γ_i meet the exceptional divisor in finitely many points. After dividing by their common exceptional factors, the pullbacks of n_1, n_2, n_3 have no common zero. Thus requiring the strict transform of V to pass through any one of these points imposes a nontrivial linear equation on the coefficients of λ_1. A general choice of v avoids all these conditions. We use the following identities from [PC26, equations (2.18)–(2.19)]:

$$m_X = \sum_{p \in \mathcal{C}} m_{\boldsymbol{n},p}^2, \qquad M_{\boldsymbol{n}} = \sum_{p \in \mathcal{C}} m_{\boldsymbol{n},p}. \tag{2.1}$$

The same choice of v, u gives

$$i_0(v, u) = m_X, \qquad 2\delta(V, 0) = m_X - M_{\boldsymbol{n}}. \tag{2.2}$$

The reduction inequalities [PC26, Propositions 2.16 and 2.17, Theorem 4.5] are

$$M_{\boldsymbol{n}} \geq m_{\boldsymbol{n}}, \qquad m_X \leq m_{\boldsymbol{n}} M_{\boldsymbol{n}}, \tag{2.3}$$

$$M_{\boldsymbol{n}} \leq m_{\mathrm{tr}}, \qquad m_X + M_{\boldsymbol{n}} \leq 3m_{\mathrm{tr}}, \tag{2.4}$$

$$\sum_i a_i(m_i - 1) \leq 2m_X - M_{\boldsymbol{n}} - b. \tag{2.5}$$

The plane curve identity $\mu = 2\delta - b + 1$ [GLS07, Proposition I.3.35], applied to $(V, 0)$, gives

$$m_X - M_{\boldsymbol{n}} = \mu(V, 0) + b - 1. \tag{2.6}$$

The polar bound (2.5) is the ramification count in Step 1 of the strategy. For the covering $u : V_s \to \overline{D}_\delta$ described there, Riemann–Hurwitz and (2.6) give total ramification $m_X + \mu(V, 0) - 1 = 2m_X - M_{\boldsymbol{n}} - b$. The a_i points of $V_s \cap \Gamma_i$ each contribute at least $m_i - 1$. Summing these contributions gives (2.5). The polar bound constrains the first two coordinates v, u. By (1.2), the third coordinate w must take different values at distinct points with the same values of v, u.

Choose reduced equations $h_i = 0$ for the curves Γ_i. The transverse multiplicities m_i along Σ_i give

$$\prod_i h_i^{m_i - 1} \mid J(n_j, n_k) \qquad (1 \leq j < k \leq 3). \tag{2.7}$$

Indeed, at a general point of Γ_i, take source coordinates (s, z) and target coordinates in which $\boldsymbol{n} = (s, a(s, z), c(s, z))$ and $\Gamma_i = \{z = 0\}$. The series $a(s, z) - a(s, 0)$ and $c(s, z) - c(s, 0)$ have z-order at least m_i, so all three Jacobian minors of $\boldsymbol{n}$ are divisible by $z^{m_i - 1}$. The orders of these minors along the irreducible curves Γ_i, together with unique factorization in B, prove (2.7).

Choose i with $m_i = m_{\mathrm{tr}}$. We next bound $a_i = i_0(v, \Gamma_i)$. This bound will imply that Γ_i is smooth when $\boldsymbol{n}$ has order three or four. The local intersection inequality and (2.5) give

$$a_i \geq m_{\boldsymbol{n}}, \qquad a_i(m_{\mathrm{tr}} - 1) \leq 2m_X - M_{\boldsymbol{n}} - b. \tag{2.8}$$

Lemma 2.9. *Suppose that $m_{\boldsymbol{n}} \in \{3,4\}$. For every index i with $m_i = m_{\mathrm{tr}}$, one has $a_i \le 4$. If $a_i = 4$, then*

$$2m_X \ge 7M_{\boldsymbol{n}} + 3b - 12. \tag{2.10}$$

Proof. If $a_i \ge 5$, then (2.4) and (2.8) give

$$5\left(\frac{m_X + M_{\boldsymbol{n}}}{3} - 1\right) \le 2m_X - M_{\boldsymbol{n}} - b,$$

and therefore, by (2.3) and $b \ge 1$,

$$\begin{aligned} 0 &\le m_X - 8M_{\boldsymbol{n}} + 15 - 3b \\ &\le (m_{\boldsymbol{n}} - 8)M_{\boldsymbol{n}} + 12 \\ &\le (m_{\boldsymbol{n}} - 8)m_{\boldsymbol{n}} + 12 < 0. \end{aligned}$$

The last expression is -3 when $m_{\boldsymbol{n}} = 3$ and -4 when $m_{\boldsymbol{n}} = 4$. Thus $a_i \le 4$. If $a_i = 4$, (2.4) and (2.8) give

$$4\left(\frac{m_X + M_{\boldsymbol{n}}}{3} - 1\right) \le 4(m_{\mathrm{tr}} - 1) \le 2m_X - M_{\boldsymbol{n}} - b,$$

which is equivalent to (2.10). □

The reduction theorem [PC26, Theorem 1.1] gives $m_{\boldsymbol{n}} \in \{2,3,4\}$.

3 Common preparation and auxiliary lemmas

We choose coordinates, bound weighted orders, and prove a criterion for noninjectivity, following the three steps of the strategy. A smooth component of the critical locus with no common tangent with V provides the axis required in Step 1. Taking this component as an axis expresses the divisibility of the Jacobian minors as divisibility by a power of its defining coordinate. The base points give lower bounds for the orders of the coefficients of a Weierstrass polynomial defining V, whose roots determine the weights for Step 2. Proposition 3.6 establishes these coordinates before we introduce the weighted estimates. For Step 3, Lemma 3.10 proves that an isolated collision of the initial forms implies noninjectivity of the analytic map.

3.1 Base points and branch counts

We call p the *predecessor* of q if q lies on the exceptional curve immediately after blowing up p. An infinitely near point q is *proximate* to p, written $q \to p$, if it lies on the strict transform of the exceptional curve created by blowing up p. A point other than the origin is *free* if it is proximate to one point, and *satellite* if it is proximate to two. For a base point $p \in \mathcal{C}$, put

$$\rho_p = m_{\boldsymbol{n},p} - \sum_{q \to p} m_{\boldsymbol{n},q}.$$

The proximity inequalities give $\rho_p \ge 0$ [CA00, Corollary 7.2.14]. For a curve germ Γ, its *path through the cluster* is the set of points $p \in \mathcal{C}$ through which its successive strict transforms pass. For $1 \le j \le m_{\boldsymbol{n}}$, put

$$A_j = \#\{p \in \mathcal{C} : m_{\boldsymbol{n},p} = j\}. \tag{3.1}$$

For coordinates (x,y) and a positive rational number α, define the weighted order

$$\nu_\alpha\left(\sum_{i,j\ge 0} h_{ij}x^i y^j\right) = \min_{h_{ij}\neq 0}(\alpha i + j), \qquad \nu_\alpha(0) = \infty.$$

The next lemma counts the branches of V, describes the base points $p \in \mathcal{C}$ with $m_{\boldsymbol{n},p} = m_{\boldsymbol{n}}$, and computes intersection multiplicities with smooth curves.

Lemma 3.2. *One has*

$$b = m_{\boldsymbol{n}} - \sum_{\substack{q \in \mathcal{C} \\ q \text{ satellite}}} m_{\boldsymbol{n},q}. \tag{3.3}$$

The base points satisfying $m_{\boldsymbol{n},p} = m_{\boldsymbol{n}}$ can be ordered as $p_0 = 0, p_1, \dots, p_r$, where each p_{j+1} lies on the exceptional curve created by blowing up p_j and on no other exceptional curve. There are coordinates in which

$$\nu_{A_{m_{\boldsymbol{n}}}}(h) \geq m_{\boldsymbol{n}} A_{m_{\boldsymbol{n}}} \qquad (h \in I). \tag{3.4}$$

Let Γ be a smooth branch having no common component with V, and suppose that their strict transforms after principalization of I do not meet over the origin. Then

$$i_0(v, \Gamma) = m_{\boldsymbol{n}} + \sum_{\substack{p \neq 0 \text{ on the path} \\ \text{followed by } \Gamma}} m_{\boldsymbol{n},p}. \tag{3.5}$$

Every point on this path other than the origin is free.

Proof. After principalization of I, the intersection number of the strict transform of V with the strict transform of the exceptional curve created at p is ρ_p. Our choice of v in Section 2 makes these intersections transverse and disjoint from crossings of the exceptional divisor. Hence $b = \sum_p \rho_p$. In this sum, each $m_{\boldsymbol{n},q}$ with $q \neq 0$ is subtracted once at the predecessor of q, and once more if q is satellite. This proves (3.3).

The integers $m_{\boldsymbol{n},p}$ do not increase along a sequence of blowups. The proximity inequality at p permits at most one point q with predecessor p and $m_{\boldsymbol{n},q} = m_{\boldsymbol{n}}$. If such a point exists, then $m_{\boldsymbol{n},p} = m_{\boldsymbol{n}}$ as well. This point q cannot be satellite. Otherwise, both q and its predecessor p would be proximate to an earlier point p', whose proximity inequality would give $m_{\boldsymbol{n},p'} \geq m_{\boldsymbol{n},p} + m_{\boldsymbol{n},q} = 2m_{\boldsymbol{n}}$. This contradicts $m_{\boldsymbol{n},p'} \leq m_{\boldsymbol{n}}$. Thus the base points p with $m_{\boldsymbol{n},p} = m_{\boldsymbol{n}}$ can be listed as $p_0 = 0, p_1, \dots, p_r$, with p_j the predecessor of p_{j+1} and each p_{j+1} free. Choose coordinates (x, y) so that the successive strict transforms of the smooth curve $x = 0$ pass through $p_0, \dots, p_r$ and $y = 0$ is transverse to $x = 0$. The successive blowup charts give $y^{-m_{\boldsymbol{n}} A_{m_{\boldsymbol{n}}}} h(y^{A_{m_{\boldsymbol{n}}}} X, y) \in \mathbb{C}\{X, y\}$ for every $h \in I$, proving (3.4). Finally, the blowup formula for intersection multiplicity, iterated along the smooth branch Γ, gives (3.5). The intersection contribution from the strict transforms of V and Γ after principalization vanishes by the disjointness assumption. After each blowup on the smooth branch Γ, its strict transform meets the new exceptional curve transversely away from the older exceptional curves. Hence every point on the path of Γ other than the origin is free. □

3.2 A common preparation for the three orders

The ramification bound (2.8) controls how many points of a nearby fiber V_s of v can lie on a component Γ_i with $m_i = m_{\mathrm{tr}}$. We now use the cluster formulas to show that this component is smooth and meets V with intersection multiplicity $m_{\boldsymbol{n}}$. This completes Step 1 and provides the coordinates for the root separation in Step 2.

We call a monic polynomial of degree $m_{\boldsymbol{n}}$ in x *centered* if its coefficient of $x^{m_{\boldsymbol{n}}-1}$ vanishes. Equivalently, its roots, counted with multiplicity, have mean zero. The next proposition produces a centered Weierstrass polynomial whose zero set is V.

Proposition 3.6 (Common preparation)**.** *Under the hypotheses of Section 2, choose i with $m_i = m_{\mathrm{tr}}$. Then Γ_i is smooth, $a_i = m_{\boldsymbol{n}}$, and its path through the base-point cluster consists only of the origin. There are coordinates (x, y) with $\Gamma_i = \{y = 0\}$ in which*

$$\begin{aligned} v = EG, \qquad G = x^{m_{\boldsymbol{n}}} + \sum_{j=0}^{m_{\boldsymbol{n}}-2} c_j(y)x^j, \qquad E \in \mathbb{C}\{x, y\}^*, \\ c_j \in \mathbb{C}\{y\}, \qquad \operatorname{ord}_y c_j \geq (m_{\boldsymbol{n}} - j)A_{m_{\boldsymbol{n}}} \quad (0 \leq j \leq m_{\boldsymbol{n}} - 2). \end{aligned} \tag{3.7}$$

If $m_{\boldsymbol{n}} = 2$ and $M_{\boldsymbol{n}} = 2$, then $m_{\mathrm{tr}} = 2$.

Proof. Write $r = \operatorname{mult}_0(\Gamma_i)$. The local intersection inequality, (2.8), and (2.4) give

$$m_{\boldsymbol{n}} r \leq a_i, \qquad a_i(m_{\mathrm{tr}} - 1) \leq 2m_X - M_{\boldsymbol{n}} - b, \qquad M_{\boldsymbol{n}} \leq m_{\mathrm{tr}}.$$

If $m_{\boldsymbol{n}} = 4$, Lemma 2.9 gives $a_i \leq 4$, hence $r = 1$ and $a_i = 4$.

Suppose $m_{\boldsymbol{n}} = 3$. The same lemma gives $r = 1$ and $a_i \in \{3, 4\}$. If $a_i = 4$, (3.5) requires one further base point p on the path of Γ_i, with p free and $m_{\boldsymbol{n},p} = 1$. This point is distinct from the satellite points counted in (3.3), so

$$3M_{\boldsymbol{n}} - m_X = 2(A_2 + A_1) \geq 2, \qquad M_{\boldsymbol{n}} \geq 3 + 1 + (3 - b) = 7 - b.$$

Consequently

$$2m_X - 7M_{\boldsymbol{n}} - 3b + 12 \leq 8 - M_{\boldsymbol{n}} - 3b \leq 1 - 2b < 0,$$

contrary to (2.10). Thus $a_i = 3$.

Now suppose $m_{\boldsymbol{n}} = 2$. The cluster formulas (2.1) and (3.3) give

$$M_{\boldsymbol{n}} = 2A_2 + A_1, \qquad m_X = 2M_{\boldsymbol{n}} - A_1, \qquad 1 \leq b \leq 2.$$

If $M_{\boldsymbol{n}} \geq 3$ and $a_i \geq 4$, then

$$4(M_{\boldsymbol{n}} - 1) \leq a_i(m_{\mathrm{tr}} - 1) \leq 3M_{\boldsymbol{n}} - b,$$

forcing $M_{\boldsymbol{n}} = 3$ and $b = 1$. But a cluster with $M_{\boldsymbol{n}} = 3$ consists of the origin, where $m_{\boldsymbol{n},0} = 2$, and one point p on the exceptional curve of the first blowup, with $m_{\boldsymbol{n},p} = 1$. Thus (3.3) gives $b = 2$. Therefore $a_i \leq 3$ and $r = 1$. If $a_i = 3$, the path of Γ_i contains a free base point p with $m_{\boldsymbol{n},p} = 1$. By (3.3), the satellite points account for $2 - b$ of the A_1 points. Including this free point gives $A_1 \geq 3 - b$. The polar bound (2.8) would then imply

$$3(M_{\boldsymbol{n}} - 1) \leq 3M_{\boldsymbol{n}} - 2A_1 - b, \qquad 3 \geq 2A_1 + b \geq 6 - b \geq 4.$$

Hence $a_i = 2$ when $M_{\boldsymbol{n}} \geq 3$.

If $M_{\boldsymbol{n}} = 2$, the origin is the only base point, so $m_X = 4$ and $b = 2$. The quadratic initial form q_v of v has two distinct roots in $\mathbb{P}^1$. Since $i_0(v, u) = 4$, the initial form q_u of u is also quadratic and has no common root with q_v. The morphism

$$[q_v : q_u] : \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad [x : y] \longmapsto [q_v(x, y) : q_u(x, y)]$$

has degree two. Riemann–Hurwitz gives two simple ramification points, neither at a zero of q_v, since q_u/q_v has simple poles there. Thus $J(q_v, q_u)$ is squarefree and coprime to q_v. Let $h_i = 0$ be an irreducible equation for Γ_i. The initial form of h_i for total degree is a nonzero multiple of L^r for a linear form L. The divisibility $h_i^{m_{\mathrm{tr}}-1} \mid J(v, u)$ from (2.7) gives

$$L^{r(m_{\mathrm{tr}}-1)} \mid J(q_v, q_u).$$

It follows that $r = 1$ and $m_{\mathrm{tr}} = 2$. Since L does not divide q_v, the curves Γ_i and V have no common tangent, and $a_i = 2$.

Thus Γ_i is smooth and $a_i = m_{\boldsymbol{n}}$ in every order. Formula (3.5) then excludes later base points on the path of Γ_i. Choose coordinates (x, y) with $\Gamma_i = \{y = 0\}$ so that the successive strict transforms of $x = 0$ pass through the base points with $m_{\boldsymbol{n},p} = m_{\boldsymbol{n}}$, ordered as in Lemma 3.2. These axes are transverse. By (3.4), Weierstrass preparation [GLS07, Theorem I.1.6] writes v as a unit times

$$x^{m_{\boldsymbol{n}}} + \sum_{j<m_{\boldsymbol{n}}} d_j(y)x^j, \qquad \operatorname{ord}_y d_j \geq (m_{\boldsymbol{n}} - j)A_{m_{\boldsymbol{n}}}.$$

The translation $x' = x + d_{m_{\boldsymbol{n}}-1}(y)/m_{\boldsymbol{n}}$ centers the roots of this Weierstrass polynomial. The added term $d_{m_{\boldsymbol{n}}-1}(y)/m_{\boldsymbol{n}}$ has order at least $A_{m_{\boldsymbol{n}}}$, so it preserves the coefficient bounds and the axis Γ_i. This proves (3.7). □

3.3 The first separation of the roots

For the formulas in this subsection, write $m = m_{\boldsymbol{n}}$. The centered polynomials supplied by Proposition 3.6 have the following forms in the three orders under consideration:

m	$G(x, y)$
2	$x^2 + c_0(y)$
3	$x^3 + c_1(y)x + c_0(y)$
4	$x^4 + c_2(y)x^2 + c_1(y)x + c_0(y)$

For Step 2, we divide the roots by powers of y and find the smallest exponent for which their limits differ as $y \to 0$. We use weighted initial forms to compute these limits and to compare the three coordinates v, u, w of $\boldsymbol{n}$. For positive integral weights (w_x, w_y), the *weighted order* of a nonzero series is the least value of $iw_x + jw_y$ among its monomials $x^i y^j$. Its *weighted initial form*, or simply *initial form*, is the sum of the monomials of that weight. We use $\operatorname{ord}_{(w_x,w_y)}$ and $\operatorname{in}_{(w_x,w_y)}$ for these operations. The same definitions apply to positive rational weights, which may be multiplied by a common denominator. A polynomial is *weighted homogeneous* if all its nonzero monomials have the same weight.

For a reduced Weierstrass polynomial G as in (3.7), define its *separation exponent* by

$$\alpha = \min_{0\leq j\leq m-2} \frac{\operatorname{ord}_y c_j}{m - j} \geq A_m, \qquad \operatorname{ord}_y 0 = \infty. \tag{3.8}$$

At least one c_j is nonzero, since otherwise $G = x^m$ would not be reduced for $m \geq 2$. Thus α is finite. By Puiseux' theorem [Wal04, Theorem 2.2.6], after a substitution $y = t^d$ for some integer $d \geq 1$, the roots of $G(x, y) = 0$ are convergent power series in t. Denote the corresponding series in fractional powers of y by $\xi_1(y), \ldots, \xi_m(y)$. These are the *Puiseux roots* of G. Normalize their orders by $\operatorname{ord}_y y = 1$. Each ξ_i has order at least α: otherwise its mth power would be the unique term of least order in $G(\xi_i(y), y) = 0$. As $y \to 0$, the substitution $x = y^\alpha z$ gives the *limiting polynomial* P_m, defined by

$$y^{-m\alpha}G(y^\alpha z, y) \longrightarrow P_m(z), \qquad \operatorname{in}_{(\alpha,1)} G = y^{m\alpha}P_m(x/y^\alpha). \tag{3.9}$$

The limits of $\xi_i(y)/y^\alpha$ are the roots of the monic polynomial P_m, counted with multiplicity. Its coefficient of z^{m-1} vanishes, and the minimum in (3.8) gives a nonzero coefficient of lower degree. If $P_m(z) = (z - \zeta)^m$ for some $\zeta \in \mathbb{C}$, its vanishing coefficient of z^{m-1} would give $\zeta = 0$, contrary to the existence of that nonzero coefficient of lower degree. Since P_m is monic of degree m, it is therefore not the mth power of a polynomial in $\mathbb{C}[z]$, and its roots are not all equal. For every rational number β with $0 < \beta < \alpha$, all rescaled roots $\xi_i(y)/y^\beta$ tend to zero as $y \to 0$. Thus α is the first exponent at which at least two rescaled roots have different limits.

3.4 Collisions of weighted initial forms

To detect noninjectivity, we study pairs of distinct points with the same image under the map formed by the weighted initial forms of v, u, w. These are the pairs whose separation is required in Step 3. Weighted homogeneity allows us to record them in a weighted projective space. The lemma below shows that an isolated collision in that space implies that the analytic map is not injective.

Use positive integral weights (w_x, w_y), clearing denominators if the given weights are rational. The weighted projective space $\mathbb{P}(w_x, w_y, w_x, w_y)$ identifies nonzero tuples (x, y, X, Y) under the rescalings

$$(x, y, X, Y) \longmapsto (\lambda^{w_x} x, \lambda^{w_y} y, \lambda^{w_x} X, \lambda^{w_y} Y), \qquad \lambda \in \mathbb{C}^*.$$

If $F_0 = (F_{1,0}, F_{2,0}, F_{3,0}) : \mathbb{C}^2 \to \mathbb{C}^3$ has nonzero weighted homogeneous polynomial components for these weights, let $\mathcal{Z}(F_0)$ be the closed algebraic set in the weighted projective space $\mathbb{P}(w_x, w_y, w_x, w_y)$ cut out by

$$F_{i,0}(x, y) - F_{i,0}(X, Y) = 0 \qquad (1 \le i \le 3).$$

Its *diagonal* is

$$\Delta = V(x - X, y - Y) \subset \mathbb{P}(w_x, w_y, w_x, w_y).$$

When only two weighted homogeneous forms P, Q are specified, $\mathcal{Z}(P, Q)$ denotes the closed algebraic set defined by $P(x, y) = P(X, Y)$ and $Q(x, y) = Q(X, Y)$ in the same weighted projective space. We call this set the *projective fiber product* of (P, Q) with itself. For a third form R, write

$$\Delta_R = V\big(R(x, y) - R(X, Y)\big).$$

Thus $\mathcal{Z}(P, Q, R) = \mathcal{Z}(P, Q) \cap \Delta_R$. We call a point of $\mathcal{Z}(P, Q, R)$ outside Δ a *collision of the initial forms*. We will use the following criterion to pass from such a point to noninjectivity of an analytic germ. For an analytic germ $F = (F_1, F_2, F_3) : (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0)$ with $F_i \neq 0$, let F_0 be the map whose components are the initial forms of the F_i for the fixed weights (w_x, w_y).

Lemma 3.10. *If $\mathcal{Z}(F_0)$ has an isolated point outside Δ, then F is not injective.*

Proof. Choose a homogeneous coordinate of weight d that is nonzero at the isolated point of $\mathcal{Z}(F_0) \setminus \Delta$. Setting this coordinate equal to one gives a finite map from $\mathbb{C}^3$ onto the corresponding chart of $\mathbb{P}(w_x, w_y, w_x, w_y)$. Indeed, this map is the quotient by the residual action of the finite group of dth roots of unity. Choose a preimage of the isolated point and local coordinates ξ_1, ξ_2, ξ_3 at that preimage. Under weighted rescaling, the equations

$$\tau^{-\operatorname{ord}_{(w_x, w_y)} F_i}\big(F_i(\tau^{w_x} x, \tau^{w_y} y) - F_i(\tau^{w_x} X, \tau^{w_y} Y)\big) = 0 \qquad (1 \le i \le 3)$$

have left-hand sides that are convergent in $(\xi_1, \xi_2, \xi_3, \tau)$ and have an isolated common zero on $\{\tau = 0\}$. Three equations in four variables have a common zero set of dimension at least one [GLS07, Theorem B.2.1(1)]. A component of positive dimension through the chosen zero is therefore not contained in $\{\tau = 0\}$. For $\tau \neq 0$, rescaling gives two distinct points of $\mathbb{C}^2$ with the same image under F. Both points approach the origin as $\tau \to 0$, proving noninjectivity. □

3.5 Weighted orders and divisibility of the Jacobian minors

We compare the separation exponent of G with the vanishing order of $J(v, u)$ along Γ_i. The following estimate uses (2.7) and (2.4) to connect Steps 1 and 2. It applies when $m_{\boldsymbol{n}} \in \{3, 4\}$. Retain the coordinates of Proposition 3.6, put $m = m_{\boldsymbol{n}} \in \{3, 4\}$, and use $\alpha, P = P_m$ from (3.8)–(3.9). By (2.7), $y^{m_{\mathrm{tr}}-1} \mid J(v, u)$. Throughout this subsection, initial forms and weighted orders are

taken with respect to $(\alpha, 1)$. Subtract terms cv^q from u while its initial form is a scalar multiple of a positive integral power of the initial form of v. These target changes leave $u|_V$ unchanged and preserve the ideals I and $(v, u)B$ and the Jacobian determinant $J(v, u)$. Each subtraction strictly increases the weighted order of the remainder. If the process did not terminate, its remainders would have arbitrarily large order at the origin, so $u \in (v) + \mathfrak{m}^N$ for every N. Krull intersection in $B/(v)$ [GLS07, Theorem B.4.2] would give $u \in (v)$, contradicting $i_0(v, u) = m_X < \infty$ in (2.2). For this choice of u, put

$$\ell = \nu_\alpha(u), \qquad \operatorname{in}_{(\alpha,1)} u = y^\ell R(x/y^\alpha), \qquad k = \deg R.$$

Define the polynomial $\Omega = \ell P'R - m\alpha PR'$. If $\Omega \neq 0$, set $j = \deg \Omega$.

Lemma 3.11. *One has $\ell = k\alpha$ and*

$$m_X \geq m_{\boldsymbol{n}}\ell. \tag{3.12}$$

If $\Omega \neq 0$, then

$$m_{\mathrm{tr}} \leq \ell + \alpha(m_{\boldsymbol{n}} - 1 - j). \tag{3.13}$$

In this case, $j \leq m_{\boldsymbol{n}} - 2$ and $k \geq m_{\boldsymbol{n}} + 1$. If $\Omega = 0$, then

$$R^{m_{\boldsymbol{n}}} = \kappa P^k \tag{3.14}$$

for some $\kappa \in \mathbb{C}^*$, and the inequality in (3.12) is strict.

Proof. Every Puiseux root $\xi(y)$ of G has order at least α, as established before (3.9). By (2.2), the intersection formula [Wal04, Lemma 1.2.1(ii)–(iii)], with roots counted with multiplicity and $\operatorname{ord}_y y = 1$, gives

$$m_X = i_0(G, u) = \sum_{G(\xi,y)=0} \operatorname{ord}_y u(\xi(y), y) \geq m\ell.$$

If $\Omega \neq 0$, differentiation gives

$$\operatorname{in}_{(\alpha,1)} J(v, u) = E(0, 0)y^{\ell+(m-1)\alpha-1}\Omega(x/y^\alpha).$$

Indeed, in $J(v, u) = EJ(G, u) + GJ(E, u)$, the summand $GJ(E, u)$ has weighted order at least $\ell + (m-1)\alpha - 1 + \min(\alpha, 1)$, so it does not contribute to the initial form of $J(v, u)$. The nonzero coefficient of z^j in $\Omega(z)$ gives a nonzero monomial $x^j y^{\ell+(m-1-j)\alpha-1}$ in that initial form. Since $y^{m_{\mathrm{tr}}-1} \mid J(v, u)$, its exponent of y is at least $m_{\mathrm{tr}} - 1$, proving (3.13). Combining (3.13) with (3.12) and (2.4) yields

$$(m-3)\ell + M_{\boldsymbol{n}} \leq 3\alpha(m - 1 - j).$$

Since $m \geq 3$, $\ell > 0$, and $M_{\boldsymbol{n}} \geq m > 0$, the quantity $(m-3)\ell + M_{\boldsymbol{n}}$ is positive. As $\alpha > 0$, this forces $m - 1 - j > 0$, hence $j \leq m - 2$. The coefficient of z^{m+k-1} in $\Omega(z)$ is $m(\ell - k\alpha)$ times the nonzero coefficient of z^k in $R(z)$. Since $m + k - 1 \geq m - 1 > j = \deg \Omega$, this coefficient vanishes, so $\ell = k\alpha$. Thus x^k occurs in u, and $u \in \mathfrak{m}^m$ gives $k \geq m$. If $k = m$, let $r_m \neq 0$ be the leading coefficient of R. Since P is monic and the choice of u in Section 3.5 excludes $R = r_m P$, the polynomial $\widetilde{R} = R - r_m P$ is nonzero and has degree $d < m$. The equality $\ell = m\alpha$ gives

$$\Omega = m\alpha(P'\widetilde{R} - P\widetilde{R}').$$

If $\widetilde{r}_d$ is the leading coefficient of $\widetilde{R}$, the coefficient of z^{m+d-1} in this expression is $m\alpha(m-d)\widetilde{r}_d \neq 0$. Hence $\deg \Omega = m + d - 1 \geq m - 1$, contradicting $j \leq m - 2$.

If $\Omega = 0$, the coefficient of z^{m+k-1} in the zero polynomial $\ell P'R - m\alpha PR'$ gives $\ell = k\alpha > 0$. The identity $kP'R - mPR' = 0$ then implies $(R^m/P^k)' = 0$ in the rational function field $\mathbb{C}(z)$. A rational function with zero derivative is constant, so $R^m/P^k = \kappa \in \mathbb{C}^*$, proving (3.14). For every Puiseux root ξ of G, the constant term ζ of ξ/y^α satisfies $P(\zeta) = 0$, and therefore $R(\zeta) = 0$. Each order $\operatorname{ord}_y u(\xi(y), y)$ in the resultant formula is then strictly greater than ℓ. □

4 Exclusion of $m_{\boldsymbol{n}} = 4$

When $m_{\boldsymbol{n}} = 4$, we compare the separation of the roots of the Weierstrass polynomial G defining V with the multiplicities of the successive strict transforms of V. In Step 2, the first separation must be into two pairs, and the multiplicities of the strict transforms of V after the corresponding blowups violate the numerical bounds. This excludes $m_{\boldsymbol{n}} = 4$ without the argument for the third coordinate in Step 3.

Theorem 4.1. *Under the hypotheses of Section 2, one has $m_{\boldsymbol{n}} \neq 4$.*

Proof. Assume $m_{\boldsymbol{n}} = 4$ and apply Proposition 3.6, with $v = EG$. Let $\alpha \geq A_4$ be the separation exponent from (3.8), and write $P = P_4$ for the limiting quartic in (3.9). Since $a_i = 4$, (2.10) holds.

To determine P, subtract scalar multiples of powers of v from u as in Section 3.5, until the initial form of u for weights $(\alpha, 1)$ is not a scalar multiple of a positive integral power of the initial form of v. Use the notation R, Ω and $k = \deg R$ from Lemma 3.11. If $\Omega \neq 0$, the lemma gives $k \geq 5$, $m_X \geq 4k\alpha$, and $m_{\mathrm{tr}} \leq (k+3)\alpha$. Also, (2.3) gives $M_{\boldsymbol{n}} \geq m_X/4$, so

$$5k\alpha \leq \frac{5}{4} m_X \leq m_X + M_{\boldsymbol{n}} \leq 3m_{\mathrm{tr}} \leq 3(k+3)\alpha,$$

contradicting $k \geq 5$.

Hence $\Omega = 0$, and (3.14) gives $R(z)^4 = \kappa P(z)^k$ in $\mathbb{C}[z]$, with $\kappa \neq 0$. Factoring both sides into linear factors shows that P and R have the same roots and, at each such root ζ,

$$4 \operatorname{ord}_{z=\zeta} R = k \operatorname{ord}_{z=\zeta} P,$$

where $\operatorname{ord}_{z=\zeta}$ denotes the order of vanishing at ζ. If k is odd, every root multiplicity of P is divisible by four, contrary to the fact that P is not a fourth power, established after (3.9). If $4 \mid k$, the identity $R^4 = \kappa P^k$ gives $R = cP^{k/4}$, contrary to the choice of u in Section 3.5. Thus $k \equiv 2 \pmod 4$, and every root multiplicity of P is even. Write $P = Q^2$ with Q a monic quadratic. The vanishing cubic coefficient of P gives

$$Q = z^2 + \eta, \qquad P = (z^2 + \eta)^2, \qquad \eta \neq 0,$$

where $\eta \neq 0$ because P is not a fourth power. Since x^k occurs in $u \in \mathfrak{m}^4$, we have $k \geq 4$, and hence $k \geq 6$. The roots of G, divided by y^α, tend in pairs to $a, -a$ with $a^2 = -\eta$. The strict inequality in (3.12) when $\Omega = 0$ gives

$$m_X > 4k\alpha \geq 24A_4. \tag{4.2}$$

We now determine the multiplicities of the strict transforms of V after this separation. The coefficient $c_2 \in \mathbb{C}\{y\}$ of x^2 in G has leading term $2\eta y^{2\alpha}$. Since $\eta \neq 0$, $2\alpha = \operatorname{ord}_y c_2$ is an integer. Since $\alpha \geq A_4$, the number $q = 2(\alpha - A_4)$ is a nonnegative integer. By Lemma 3.2, the A_4 base points p with $m_{\boldsymbol{n},p} = 4$ form a sequence in which each point after the origin lies on the exceptional curve created by blowing up the preceding point. After blowing up these points in order, the strict transform of V is defined by the polynomial

$$G_{A_4}(X, y) = y^{-4A_4} G(y^{A_4} X, y) = X^4 + \widetilde{c}_2(y) X^2 + \widetilde{c}_1(y) X + \widetilde{c}_0(y),$$

whose coefficients satisfy

$$\operatorname{ord}_y \widetilde{c}_2 = q, \qquad \operatorname{ord}_y \widetilde{c}_1 > \tfrac{3}{2} q, \qquad \operatorname{ord}_y \widetilde{c}_0 = 2q.$$

If $q \geq 2$, every monomial of G_{A_4} has ordinary total degree at least four, and X^4 occurs with coefficient one. Thus the strict transform would have multiplicity exactly four at $(0, 0)$. By

Section 2, this would be a further base point p with $m_{\boldsymbol{n},p} = 4$, contradicting the definition of A_4 in (3.1). If $q = 0$, the strict transform meets the exceptional curve $y = 0$ at $(a, 0)$ and $(-a, 0)$, since $G_{A_4}(X, 0) = (X^2 + \eta)^2$ and $a^2 = -\eta \neq 0$. At either point, the restriction of G_{A_4} to $y = 0$ has order two in $X - a$ or $X + a$, respectively. The multiplicity of the plane curve germ $\{G_{A_4} = 0\}$ at that point is therefore at most two. If $q = 1$, the restriction $G_{A_4}(X, 0) = X^4$ gives only the point $(0, 0)$ on the exceptional curve. The lowest homogeneous part of G_{A_4} for the ordinary total degree is $\eta^2 y^2 \neq 0$, so the plane curve germ has multiplicity two there. Every remaining base point maps to one of the points just described under the intervening blowups. Every Puiseux root $\xi(y)$ of G has order at least $\alpha \geq A_4$. Hence $\xi(y)/y^r$ remains bounded for $1 \leq r \leq A_4$ and tends to zero for $r < A_4$. Thus the successive strict transforms of every branch of V lie in the charts with coordinates $(x/y^r, y)$, and after blowing up all A_4 points they pass through one of the points just listed. By [Wal04, Lemma 3.4.2] and the local intersection inequality, multiplicity cannot increase under further blowups. Since the multiplicities of the strict transforms of V at the base points are $m_{\boldsymbol{n},p}$, all remaining base points satisfy $m_{\boldsymbol{n},p} \leq 2$. Therefore (2.1) gives

$$m_X \leq 16A_4 + 2(M_{\boldsymbol{n}} - 4A_4) = 2M_{\boldsymbol{n}} + 8A_4. \tag{4.3}$$

Combining (4.3) with (4.2) gives $M_{\boldsymbol{n}} > 8A_4 \geq 8$ and $m_X < 3M_{\boldsymbol{n}}$. Thus the average $m_X/M_{\boldsymbol{n}}$ of the multiplicities $m_{\boldsymbol{n},p}$, with weights proportional to $m_{\boldsymbol{n},p}$, is below three. However, integrality gives $M_{\boldsymbol{n}} \geq 9$, and the polar estimate (2.10) gives

$$2m_X \geq 7M_{\boldsymbol{n}} + 3b - 12 \geq 7M_{\boldsymbol{n}} - 9 \geq 6M_{\boldsymbol{n}},$$

a contradiction. □

5 Exclusion of $m_{\boldsymbol{n}} = 3$

When $m_{\boldsymbol{n}} = 3$, the numerical bounds determine two possible pairs of weighted initial forms of v, u in Step 2. For Step 3, the projective fiber product records pairs of points with the same image under the map defined by each pair of forms, up to simultaneous rescaling. The divisibility conditions on the Jacobian minors prevent the third coordinate from separating all such pairs. The isolated collisions obtained from these initial forms lift to the analytic map $\boldsymbol{n}$ and contradict (1.2).

In the next two subsections, assume $m_{\boldsymbol{n}} = 3$ and use the component Γ_i and coordinates of Proposition 3.6. Thus $\Gamma_i = \{y = 0\}$ and $a_i = 3$. Put $a = A_3 \geq 1$. Since $m_{\boldsymbol{n},p} \in \{1, 2, 3\}$ at every base point $p \in \mathcal{C}$, (2.1) and (3.3) give

$$2M_{\boldsymbol{n}} - m_X = \sum_{p \in \mathcal{C}} m_{\boldsymbol{n},p}(2 - m_{\boldsymbol{n},p}) \geq -3a, \qquad b \leq 3. \tag{5.1}$$

5.1 The two possible pairs of weighted initial forms

In independent variables x, t, define the pairs of homogeneous polynomials

$$E: \quad P = x^3 + 3t^3, \qquad Q = x^4 + 4xt^3, \tag{5.2}$$

$$H: \quad P = x^3 + 3xt^2, \qquad Q = x^4 + 4x^2t^2 + 2t^4. \tag{5.3}$$

The Jacobian of each pair is

$$J_{x,t}(P, Q) = \begin{cases} -36t^5, & E, \\ 24t^5, & H. \end{cases} \tag{5.4}$$

The following proposition identifies E and H, after substituting $t = y^a$, as the possible pairs of initial forms of v, u for weights $(a, 1)$ and determines the transverse multiplicity m_{tr}.

Proposition 5.5. *Under the assumptions of Section 5, there are source coordinates (x, y) with $\Gamma_i = \{y = 0\}$ and changes of target coordinates, obtained by subtracting a polynomial in v from u and multiplying v, u by nonzero constants, such that*

$$\operatorname{in}_{(a,1)} v = P(x, y^a), \qquad \operatorname{in}_{(a,1)} u = Q(x, y^a)$$

for one of the pairs (5.2), (5.3). Moreover,

$$m_X = 12a, \qquad m_{\mathrm{tr}} = 6a. \tag{5.6}$$

Proof. Let $\alpha \geq a$ be the separation exponent of G from (3.8), and write the limiting cubic as

$$p = P_3 = z^3 + a_p z + b_p.$$

As established after (3.9), the monic cubic $p \in \mathbb{C}[z]$ is not equal to $(z - c)^3$ for any $c \in \mathbb{C}$. Its root multiplicities sum to three, so at least one root is simple. Apply Lemma 3.11 to (v, u) and retain its notation R, Ω, ℓ, k. If $\Omega = 0$, the identity $R^3 = \kappa p^k$ in $\mathbb{C}[z]$ gives $3 \operatorname{ord}_{z=c} R = k$ at a simple root c of p, so $3 \mid k$. At every root of p, the multiplicity as a root of R is $k/3$ times the multiplicity as a root of p, and R has no other roots. Thus $R = c_0 p^{k/3}$ for some $c_0 \in \mathbb{C}^*$. Since $\ell = k\alpha$, the initial form of u for weights $(\alpha, 1)$ would then be a scalar multiple of the $(k/3)$rd power of that of v, contrary to the choice of u in Section 3.5. Thus $\Omega \neq 0$. With $j = \deg \Omega$ and $k = \deg R$, the lemma gives

$$j \leq 1, \quad k \geq 4, \quad \ell = k\alpha, \quad m_X \geq 3k\alpha, \quad m_{\mathrm{tr}} \leq \alpha(k + 2 - j).$$

By (5.1) and (2.4),

$$\frac{9}{2} k\alpha - \frac{3}{2} a \leq m_X + M_{\boldsymbol{n}} \leq 3m_{\mathrm{tr}} \leq 3\alpha(k + 2 - j).$$

Hence $(k - 4 + 2j)\alpha \leq a \leq \alpha$. Dividing by $\alpha > 0$ gives $k - 4 + 2j \leq 1$. Since $k \geq 4$ and $j \geq 0$ are integers, this forces $j = 0$. At every root c of p, $0 \neq \Omega(c) = \ell p'(c) R(c)$. Hence $p'(c) \neq 0$ and $R(c) \neq 0$ at every root of p. Thus all roots of p are simple, and p, R are coprime in $\mathbb{C}[z]$. Along each of the three Puiseux roots of G, the orders of the restrictions of u and G_x, normalized by $\operatorname{ord}_y y = 1$, are therefore exactly $k\alpha$ and 2α, respectively. Summing these orders in the resultant formulas for $i_0(G, u) = m_X$ and $i_0(G, G_x)$ gives

$$m_X = 3k\alpha, \qquad i_0(G, G_x) = 6\alpha.$$

Since $v = EG$ and E is a unit, $v_x \equiv EG_x \pmod{v}$ gives $i_0(v, v_x) = i_0(G, G_x) = 6\alpha$. The axis $y = 0$ is transverse to V, so $i_0(v, y) = 3$. The polar identity [Wal04, Lemma 7.2.1 and its consequence, pp. 160–161] is

$$i_0(v, v_x) = \mu(V, 0) + i_0(v, y) - 1 = \mu(V, 0) + 2.$$

Thus $\mu(V, 0) = 6\alpha - 2$, and (2.6) yields

$$M_{\boldsymbol{n}} = 3\alpha(k - 2) - b + 3.$$

Substituting $m_X = 3k\alpha$ and $M_{\boldsymbol{n}} = 3\alpha(k - 2) - b + 3$ into $m_X + M_{\boldsymbol{n}} \leq 3m_{\mathrm{tr}} \leq 3\alpha(k + 2)$ gives $3\alpha(k - 4) \leq b - 3$. Since $\alpha \geq a \geq 1$, $k \geq 4$, and (3.3) gives $b \leq 3$, we have $0 \leq 3\alpha(k - 4) \leq b - 3 \leq 0$, hence $k = 4$ and $b = 3$. Substituting these values into the formulas for $m_X, M_{\boldsymbol{n}}$ and using $M_{\boldsymbol{n}} \leq m_{\mathrm{tr}} \leq \alpha(k + 2)$ gives

$$k = 4, \qquad b = 3, \qquad m_X = 12\alpha, \qquad M_{\boldsymbol{n}} = m_{\mathrm{tr}} = 6\alpha.$$

The multiplicities of the three branches of V sum to $\operatorname{mult}_0 V = 3$, so each branch is smooth. Transversality to $y = 0$ gives three convergent graphs $x = \xi_s(y)$, and the limits of $\xi_s(y)/y^\alpha$ as $y \to 0$

are the distinct roots c_s of p. At least one c_s is nonzero, so the corresponding series $\xi_s(y)$ has order α, proving $\alpha \in \mathbb{Z}$. For $0 \le r < \alpha$, the strict transforms of all three branches under the substitutions $x = y^r x_r$ pass through $(x_r, y) = (0, 0)$. After the αth blowup, they meet the exceptional curve at the distinct points $(c_s, 0)$. Thus the strict transforms of all three branches pass through exactly α common centres of blowup, including the origin. At each centre the strict transform of V has multiplicity three, so each centre is a base point p with $m_{\boldsymbol{n},p} = 3$. Therefore $a \ge \alpha$, so $a = \alpha$, proving (5.6).

It remains to classify the initial forms of v, u for weights $(a, 1)$. Set $\chi = R$ and write $P = t^3 p(x/t)$, $Q = t^4 \chi(x/t)$. Since $\ell = 4\alpha$, the nonzero constant Ω satisfies $\Omega = \alpha(4p'\chi - 3p\chi')$. Put $c = \Omega/\alpha \neq 0$, so $4p'\chi - 3p\chi' = c$. Comparing coefficients of powers of z in this polynomial identity, with c_4 the coefficient of z^4 in $\chi(z)$, gives

$$\chi = c_4 \left(z^4 + \frac{4a_p}{3} z^2 + \frac{4b_p}{3} z + \frac{2a_p^2}{9} \right), \qquad a_p b_p = 0, \qquad c = c_4 \left(\frac{8a_p^3}{9} - 4b_p^2 \right).$$

The identity $a_p b_p = 0$ implies that at least one of a_p, b_p is zero, and $c \neq 0$ excludes $a_p = b_p = 0$. Thus exactly one is zero. Rescaling x, v, u makes the nonzero member of $\{a_p, b_p\}$ equal to three and both polynomials P, Q monic in x, giving (5.2) or (5.3). □

In these coordinates, (2.7) and (5.6) imply

$$y^{6a-1} \mid J(n_j, n_\ell) \qquad (1 \le j < \ell \le 3). \tag{5.7}$$

The chain rule preserves this divisibility when a polynomial in v, u is subtracted from the third coordinate, as in the proof of Theorem 5.17.

5.2 Collisions of the pairs of homogeneous forms

For (5.2) put $q = 3$, and for (5.3) put $q = 2$. The transformations $(x, t) \mapsto (x, \xi t)$, with $\xi^q = 1$, define the *cyclic action* used below. A polynomial R is *invariant* under this action if $R(x, \xi t) = R(x, t)$ for every $\xi^q = 1$. In coordinates $[x : t : X : T]$ on $\mathbb{P}^3$, define the lines

$$L_\xi = \{(X, T) = (x, \xi t)\} \qquad (\xi^q = 1).$$

The line L_1 is the diagonal. For each pair (P, Q), the projective fiber product $\mathcal{Z}(P, Q)$ consists of these lines and the inverse image of a curve under the substitutions $w = t^q$ and $\widetilde{w} = T^q$. The points of this fiber product outside the diagonal represent pairs of distinct points with equal values of P, Q, as in Step 3. A collision of the three forms P, Q, R may lie on one of the lines L_ξ. If the third form R is invariant under this action, all three forms are polynomials in x, w, where $w = t^q$. In case H, they take equal values at $(0, w), (0, -w)$ with $w \neq 0$ when the degree of R in x, t is odd, and at $(x, -x^2/3), (-x, -x^2/3)$ with $x \neq 0$ when that degree is even. In case E, the derivative identity (5.14) forces a collision. For the remaining forms R, we compare $R(x, t) - R(X, T)$ with $t^{\deg R} - T^{\deg R}$. The map defined by $P, Q, t^{\deg R}$ has a collision, and the comparison prevents all zeros of $R(x, t) - R(X, T)$ on the remaining curve components of $\mathcal{Z}(P, Q)$ from lying on the diagonal. Put

$$\Delta = V(x - X, t - T), \qquad p_0 = [1 : 0 : 1 : 0], \qquad \Delta_R = V\big(R(x, t) - R(X, T)\big).$$

Proposition 5.8. *Let (P, Q) be either pair (5.2), (5.3), and let $R \in \mathbb{C}[x, t]$ be homogeneous. If*

$$R \notin \mathbb{C}[P, Q], \qquad t^5 \mid J_{x,t}(P, R), \qquad t^5 \mid J_{x,t}(Q, R), \tag{5.9}$$

then $\mathcal{Z}(P, Q, R)$ has an isolated point outside the diagonal.

Proof. Put $N = \deg R \geq 1$. Set $w = t^q$, $\widetilde{w} = T^q$, and write $P = \overline{P}(x,w)$, $Q = \overline{Q}(x,w)$. In each case, the map

$$\phi : \mathbb{C}^2_{x,w} \longrightarrow \mathbb{C}^2, \qquad (x,w) \longmapsto (\overline{P}(x,w), \overline{Q}(x,w))$$

is finite because its components $\overline{P}, \overline{Q}$, homogeneous for weights $\deg x = 1$ and $\deg w = q$, have only the origin as a common zero. Indeed, on $P = 0$, the form Q equals xt^3 in case E, and equals $2t^4$ or $-t^4$ in case H. The Jacobian of ϕ is

$$J_{x,w}(\overline{P}, \overline{Q}) = \begin{cases} -12w, & E, \\ 12w^2, & H. \end{cases}$$

Components of $\mathcal{Z}(P,Q)$. In coordinates $[x : w : X : \widetilde{w}]$ on $\mathbb{P}(1,q,1,q)$, let D be the closure of the projective classes represented by pairs of distinct points $(x,w), (X,\widetilde{w})$ in a fiber of ϕ. The map

$$\rho_q : \mathbb{P}^3 \longrightarrow \mathbb{P}(1,q,1,q), \qquad [x : t : X : T] \longmapsto [x : t^q : X : T^q]$$

is well defined by the coordinate weights and finite because the corresponding extension of homogeneous coordinate rings adjoins q-th roots of $w, \widetilde{w}$. We consider every curve component of $\rho_q^{-1}(D)$. If $R(x,t) - R(X,T)$ is not identically zero on any component, its zeros on their union are isolated.

For E, the equations for D are

$$w = \frac{(X-x)(X^2+2Xx+3x^2)}{12}, \qquad \widetilde{w} = -\frac{(X-x)(3X^2+2Xx+x^2)}{12}.$$

These equations define an isomorphism $\mathbb{P}^1_{[x:X]} \to D$, whose inverse is the projection of D to $[x : X]$, since x, X cannot both vanish on D. On $x = 1$, put $\lambda = X$. Then

$$w = \frac{(\lambda-1)(\lambda^2+2\lambda+3)}{12}, \qquad \widetilde{w} = -\frac{(\lambda-1)(3\lambda^2+2\lambda+1)}{12}. \tag{5.10}$$

For H, on the chart $x = 1$ with $X \neq 0$, the fiber equation $\overline{P}(1,w) = \overline{P}(X,\widetilde{w})$ gives

$$\widetilde{w} = \frac{1+3w-X^3}{3X}. \tag{5.11}$$

Substitution in the fiber equation $\overline{Q}(1,w) = \overline{Q}(X,\widetilde{w})$ gives $(X-1)\Gamma_H(X,w)/(9X^2) = 0$, where

$$\Gamma_H = 18(X+1)w^2 - 12(X-1)(2X+1)w + (X-1)^2(X^3+3X^2+6X+2). \tag{5.12}$$

After eliminating $\widetilde{w}$ by (5.11), the factor $X-1$ defines the diagonal, and $\Gamma_H = 0$ defines D in the (X,w) chart. Regarded as a quadratic in w over $\mathbb{C}(X)$, Γ_H has discriminant $-72X^2(X-1)^2(X^2+4X+1)$, which is not a square in $\mathbb{C}(X)$ because X^2+4X+1 has two simple roots, neither equal to 0 or 1. Hence Γ_H is irreducible in $\mathbb{C}(X)[w]$. Its coefficients have no common nonconstant factor in $\mathbb{C}[X]$, so Gauss's lemma implies that it is irreducible in $\mathbb{C}[X,w]$ as well. Thus D is irreducible.

Let C be the reduced union of the curve components of $\rho_q^{-1}(D)$. Each component maps onto D, since ρ_q is finite and D is irreducible. The point $\rho_q(p_0)$ lies on D, and its inverse image is $\{p_0\}$, so every component of C passes through p_0. Consequently, the fiber product decomposes as

$$\big|\mathcal{Z}(P,Q)\big| = C \cup \bigcup_{\xi^q=1} L_\xi, \qquad L_1 = \Delta. \tag{5.13}$$

No further curve component of $\mathcal{Z}(P,Q)$ is contained in $xX = 0$. Indeed, neither coordinate pair (x,t) nor (X,T) is zero at a point of $\mathcal{Z}(P,Q)$, and finiteness of (P,Q) gives only finitely many

points of $\mathcal{Z}(P,Q)$ over any fixed $[x:t]$ or $[X:T]$. The affine fiber product of the finite map (P,Q) with itself is finite over either copy of $\mathbb{C}^2$, so it has dimension two. The ideal generated by its two defining differences $P(x,t)-P(X,T)$ and $Q(x,t)-Q(X,T)$ therefore has height two in $\mathbb{C}[x,t,X,T]$. The projectivization $\mathcal{Z}(P,Q)$ of this affine fiber product has pure dimension one. Distinct lines L_ξ meet only at p_0, and C meets their union only there. Indeed, ϕ is locally biholomorphic for $w \neq 0$, so D cannot meet the diagonal $\{x = X, w = \widetilde{w}\}$ there. On that diagonal, $w = 0$ represents only $\rho_q(p_0)$.

Forms R invariant under the cyclic action. Choose a primitive q-th root of unity ξ. If the restriction of $R(x,t) - R(X,T)$ to L_ξ in (5.13) is not identically zero and has a zero away from p_0, that zero is an isolated collision. First suppose that the restriction vanishes identically. Then $R = R_0(x,w)$ with $R_0 \in \mathbb{C}[x,w]$ of weighted degree N for $\deg x = 1$, $\deg w = q$. The difference $R_0(x,w) - R_0(X,\widetilde{w})$ is not identically zero on D. Otherwise R_0 would be constant on the fibers of ϕ over its regular values. The finite extension $\mathbb{C}(\overline{P},\overline{Q}) \subset \mathbb{C}(x,w)$ is separable. Constancy on these fibers would make all conjugates of R_0 over $\mathbb{C}(\overline{P},\overline{Q})$ coincide, so its minimal polynomial over that field would have degree one and $R_0 \in \mathbb{C}(\overline{P},\overline{Q})$. Finiteness of ϕ also makes R_0 integral over the polynomial ring $\mathbb{C}[\overline{P},\overline{Q}]$. This ring is integrally closed, so $R_0 \in \mathbb{C}[\overline{P},\overline{Q}]$. Substituting $w = t^q$ would give $R \in \mathbb{C}[P,Q]$, contrary to (5.9). Since every component of C maps onto D, none is contained in Δ_R. Any zero of $R_0(x,w) - R_0(X,\widetilde{w})$ on D at a pair of distinct points gives, after taking q-th roots of $w, \widetilde{w}$, a point of $C \cap \Delta_R$ away from p_0. By (5.13), such a point is an isolated collision.

In case H, weighted homogeneity of R_0 gives such a pair as follows. If N is odd, every monomial of R_0 contains x. Thus $(0,w)$ and $(0,-w)$ have the same values of $\overline{P}, \overline{Q}, R_0$ for $w \neq 0$. If N is even, the exponent of x in every monomial of R_0 is even. The points $(x, -x^2/3)$ and $(-x, -x^2/3)$ then have the same three values for $x \neq 0$.

In case E, use (5.10) to define the polynomial

$$F(\lambda) = R_0(1, w(\lambda)) - R_0(\lambda, \widetilde{w}(\lambda)).$$

The polynomial F is nonzero and satisfies $F(1) = 0$. The chain rule and (5.9) give

$$w \mid J_{x,w}(\overline{P}, R_0), \qquad w \mid J_{x,w}(\overline{Q}, R_0).$$

Define the polynomials

$$\Lambda_P = \frac{J_{x,w}(\overline{P}, R_0)}{w}, \qquad \Lambda_Q = \frac{J_{x,w}(\overline{Q}, R_0)}{w}.$$

Since $J_{x,w}(\overline{P},\overline{Q}) = -12w$, Cramer's rule gives

$$dR_0 = \frac{\Lambda_Q}{12}\, d\overline{P} - \frac{\Lambda_P}{12}\, d\overline{Q}.$$

For a polynomial $\Phi \in \mathbb{C}[x,w]$, write $[\Phi] = \Phi(1, w(\lambda)) - \Phi(\lambda, \widetilde{w}(\lambda))$. On the chart $x = 1$ of D, the common values of $\overline{P}$ and $\overline{Q}$ are $1 + 3w(\lambda)$ and $1 + 4w(\lambda)$. Pulling back the identity for dR_0 along the maps from $\mathbb{C}$ to $\mathbb{C}^2$ given by $\lambda \mapsto (1, w(\lambda))$ and $\lambda \mapsto (\lambda, \widetilde{w}(\lambda))$, and subtracting, yields

$$F'(\lambda) = \frac{w'(\lambda)}{12}(3[\Lambda_Q] - 4[\Lambda_P]). \tag{5.14}$$

By (5.14), $w'(\lambda) = (3\lambda^2 + 2\lambda + 1)/12$ divides $F'(\lambda)$, and the roots of $w'(\lambda)$ are different from 1. If 1 were the only zero of F, then $F = c(\lambda - 1)^k$ with $c \neq 0$ and $k \geq 1$, contradicting this divisibility. Hence F has a zero with $\lambda \neq 1$, which gives a pair of distinct points with equal values of $\overline{P}, \overline{Q}, R_0$.

The remaining forms R. It remains to consider a nonzero restriction of $R(x,t) - R(X,T)$ to L_ξ whose only zero is p_0. This restriction is a multiple of t^N. Comparison of powers of t then gives

$$R(x,t) = R_0(x,t^q) + ct^N, \qquad c \neq 0, \qquad q \nmid N.$$

Here $R_0 \in \mathbb{C}[x,w]$. In case E, (5.10) gives one smooth branch of D at $\rho_q(p_0)$, with w a local parameter and $\widetilde{w}/w \to -1$. In case H, the tangent cone of the curve $\Gamma_H = 0$, with Γ_H defined in (5.12), at $(X,w) = (1,0)$ is defined by the quadratic form

$$12\big((X-1)^2 - 3(X-1)w + 3w^2\big).$$

The two distinct lines of this tangent cone give two smooth branches of D, $X = 1 + \lambda w + O(w^2)$, where $\lambda^2 - 3\lambda + 3 = 0$. By (5.11), the corresponding limits of $\widetilde{w}/w$ are $1 - \lambda$. Thus w is a local parameter on every branch of D at $\rho_q(p_0)$, and all these limits are nonzero and different from 1. Taking q-th roots of $w, \widetilde{w}$ gives, on every local branch of C at p_0,

$$w = t^q, \qquad T = tU(t^q), \qquad U \in \mathbb{C}\{w\}^*.$$

We compare $R(x,t) - R(X,T)$ with $t^N - T^N$. The latter vanishes at the following points of C:

	X	$t = T$	x
E	ωx	1	$x^3 = -4, \quad \omega^3 = 1,\ \omega \neq 1$
H	$-x$	1	$x^2 = -3.$

(5.15)

For E, the values of P agree and both values of Q vanish. For H, both values of P vanish and the values of Q agree. At the images of these points under ρ_q, the coordinate pairs (x,w) and $(X,\widetilde{w})$ are distinct, so the points lie on C away from all the lines L_ξ. The rational function $\widetilde{w}/w$ on D equals 1 at their images, whereas its limits at $\rho_q(p_0)$ differ from 1. It is therefore nonconstant. If $t^N - T^N$ vanished on a component of C, then T/t would be a constant Nth root of unity there. Since that component maps onto D, the function $\widetilde{w}/w = (T/t)^q$ would be constant on D, a contradiction. Thus no component of C is contained in Δ_{t^N}.

On each local branch of C at p_0, in the chart $x = 1$, the difference $R(x,t) - R(X,T)$ is

$$A(t^q) + ct^N\big(1 - U(t^q)^N\big), \qquad A \in \mathbb{C}\{w\}.$$

Here $A(w)$ is the restriction of $R_0(x,w) - R_0(X,\widetilde{w})$ to the corresponding local branch of D. The exponents in the two summands have different residues modulo q, so their leading terms cannot cancel. Hence

$$\operatorname{ord}_t\big(R(x,t) - R(X,T)\big) \leq \operatorname{ord}_t(t^N - T^N).$$

Since every component of C passes through p_0, the inequality implies that none is contained in Δ_R. For a hypersurface $\mathcal{H}$ containing no component of C, we write $i_p(C,\mathcal{H})$ for the sum of the vanishing orders of a local equation of $\mathcal{H}$ along the normalized branches of C over p. We write $\deg C$ for the number of intersections of C with a general hyperplane. Both differences have degree N. The curve–hypersurface form of Bézout's theorem, together with the zero in (5.15) away from p_0, gives

$$i_{p_0}(C,\Delta_R) \leq i_{p_0}(C,\Delta_{t^N}) < N \deg C.$$

Bézout's theorem gives total intersection number $N \deg C$ for C and Δ_R, so the strict inequality leaves an intersection point away from p_0. Since no component of C is contained in Δ_R, this point is isolated in $C \cap \Delta_R$. By (5.13), C meets the diagonal and all remaining components of $\mathcal{Z}(P,Q)$ only at p_0. The point is therefore outside the diagonal and isolated in $\mathcal{Z}(P,Q,R)$. □

5.3 Exclusion of order three

The following lemma prepares the third coordinate when $m_{\boldsymbol{n}} = 3$ and when $m_{\boldsymbol{n}} = 2$. Once the initial forms of the first two coordinates define a finite map, we subtract polynomials in the first two coordinates from the third coordinate, as in Step 3. On a fiber of the first two coordinates, each subtracted function is constant, so every difference of values of the third coordinate is preserved. When the map given by the first two coordinates has degree greater than one, the lemma shows that injectivity forces this process to reach an initial form of the third coordinate outside the polynomial algebra generated by the first two initial forms.

For fixed weights (w_x, w_y), let $G, U \in \mathfrak{m}$ have initial forms P, Q with no common zero except the origin, and put $A = \mathbb{C}\{G, U\}$. Give A the filtration induced by weighted order on B, and write $\operatorname{gr} A$ for its *associated graded ring*. We identify it with the subalgebra of $\mathbb{C}[x, y]$ generated by the initial forms of elements of A. Let $H \in B$.

Lemma 5.16. *The map germ $(G, U) : (\mathbb{C}^2, 0) \to (\mathbb{C}^2, 0)$ is finite, and its fibers over regular values consist of $i_0(G, U)$ distinct points. Moreover, $\operatorname{gr} A = \mathbb{C}[P, Q]$. If $i_0(G, U) > 1$ and $H \in A$, then (G, U, H) is not injective. If $H \notin A$, successive subtraction from H of polynomials in G, U that cancel the initial form of the current remainder terminates at a remainder whose initial form lies outside $\mathbb{C}[P, Q]$.*

Proof. Suppose that nonzero common zeros of G, U approach the origin. For each such zero (x, y), set $t = \max\{|x|^{1/w_x}, |y|^{1/w_y}\}$. Rescaling to $(x/t^{w_x}, y/t^{w_y})$ and taking a subsequence would give a nonzero common zero of P, Q. Thus (G, U) is finite by [GLS07, Theorem I.1.66]. The substitution homomorphism $\mathbb{C}\{S, T\} \to B$, $F \mapsto F(G, U)$, makes B a finitely generated module over its image A, so $\dim A = \dim B = 2$. Its kernel is a prime ideal of height zero in the domain $\mathbb{C}\{S, T\}$, and is therefore zero. Thus $A \cong \mathbb{C}\{S, T\}$. Both A and B are regular local rings of dimension two, and G, U form a B-regular sequence. Hence B is free over A by [GLS07, Theorem B.8.11 and Proposition B.3.5], of rank $\dim_{\mathbb{C}} B/(G, U) = i_0(G, U)$. In characteristic zero the fibers of (G, U) over regular values have this many distinct points [GLS07, Theorem I.1.81(2)]. A function in A is constant on these fibers, proving that (G, U, H) is not injective when $i_0(G, U) > 1$ and $H \in A$.

The finite polynomial map (P, Q) makes $\mathbb{C}[x, y]$ integral over $\mathbb{C}[P, Q]$. Hence $\mathbb{C}[P, Q]$ has dimension two, and its two generators P, Q are algebraically independent. Give S, T weights equal to the weighted orders of G, U, respectively. For a nonzero $F \in \mathbb{C}\{S, T\}$, let F_d be its initial form for these weights. Algebraic independence gives $F_d(P, Q) \neq 0$, so the initial form of $F(G, U)$ is precisely $F_d(P, Q)$. This proves $\operatorname{gr} A = \mathbb{C}[P, Q]$. Each cancellation strictly increases the weighted order of the remainder. Suppose that the process did not terminate. With $\mathfrak{m}_A = (G, U)A$, the ideal $\mathfrak{m}_A B = (G, U)B$ is $\mathfrak{m}$-primary, so $\mathfrak{m}^c \subset \mathfrak{m}_A B$ for some $c \geq 1$. Since the weighted orders of the remainders are unbounded, $H \in A + \mathfrak{m}^{cn} \subset A + \mathfrak{m}_A^n B$ for every n. The class of H in the finite A-module B/A then vanishes by the Krull intersection theorem [GLS07, Theorem B.4.2], contradicting $H \notin A$. □

Applied to v, u with weights $(a, 1)$, Lemma 5.16 supplies an initial form of the third coordinate of $\boldsymbol{n}$ outside the algebra generated by the initial forms of v, u. If this form is a polynomial in x, y^a, Proposition 5.8 applies after the substitution $t = y^a$. Otherwise, multiplying y by an a-th root of unity or choosing the points in (5.15) gives a collision.

Theorem 5.17. *Under the hypotheses of Section 2, one has $m_{\boldsymbol{n}} \neq 3$.*

Proof. Assume $m_{\boldsymbol{n}} = 3$, and use the coordinates of Proposition 5.5. Put $\widehat{P} = P(x, y^a)$, $\widehat{Q} = Q(x, y^a)$, and denote the third coordinate by w, so $\boldsymbol{n} = (v, u, w)$. The map $(\widehat{P}, \widehat{Q})$ is finite. By Lemma 5.16,

(v,u) has degree $m_X = 12a > 1$. Injectivity of $\boldsymbol{n}$ therefore implies $w \notin \mathbb{C}\{v,u\}$. By Lemma 5.16, subtracting a polynomial in v,u from w gives an initial form

$$\mathcal{R} = \mathrm{in}_{(a,1)}\, w \notin \mathbb{C}[\widehat{P},\widehat{Q}].$$

Write the weight of $\mathcal{R}$ as $aN + r$, with $0 \le r < a$. Then $\mathcal{R} = y^r S(x,y^a)$, where $S(x,t)$ is homogeneous of degree N.

If $r = 0$, taking initial forms with weights $(a,1)$ in (5.7) and using

$$J_{x,y}(P(x,y^a),S(x,y^a)) = ay^{a-1}J_{x,t}(P,S)(x,y^a)$$

gives $t^5 \mid J_{x,t}(P,S), J_{x,t}(Q,S)$, including when either Jacobian is zero. Since $S \notin \mathbb{C}[P,Q]$, Proposition 5.8 gives an isolated non-diagonal point of $\mathcal{Z}(P,Q,S)$. The map

$$\rho_a : \mathbb{P}(a,1,a,1) \longrightarrow \mathbb{P}^3, \qquad [x:y:X:Y] \longmapsto [x:y^a:X:Y^a]$$

is well defined, since its coordinate polynomials have weighted degree a. It is finite because the corresponding extension of homogeneous coordinate rings adjoins a-th roots of t,T, and surjective because these roots exist over $\mathbb{C}$. Substituting $t = y^a$ and $T = Y^a$ in the equations defining $\mathcal{Z}(P,Q,S)$ gives $\mathcal{Z}(\widehat{P},\widehat{Q},\mathcal{R}) = \rho_a^{-1}(\mathcal{Z}(P,Q,S))$. The inverse image under ρ_a of the isolated collision is therefore nonempty and consists of isolated points. These points lie outside the diagonal in $\mathbb{P}(a,1,a,1)$, since ρ_a maps this diagonal into the diagonal in $\mathbb{P}^3$.

Suppose $0 < r < a$, and choose a primitive a-th root of unity ζ. On $(X,Y) = (x,\zeta y)$, the differences of the values of $\widehat{P}$ and $\widehat{Q}$ vanish, and $\mathcal{R}(x,y) - \mathcal{R}(X,Y)$ is $(1-\zeta^r)y^r S(x,y^a)$. If S has a zero $[x:t] \in \mathbb{P}^1$ with $t \neq 0$, that zero gives a collision of $(\widehat{P},\widehat{Q},\mathcal{R})$ with $y \neq 0$. It is isolated: by (5.4), the differences for $\widehat{P},\widehat{Q}$ locally define the curve $(X,Y) = (x,\zeta y)$. On this curve in the chart $y = 1$, the difference for $\mathcal{R}$ restricts to a nonzero multiple of $S(x,1)$.

If S has no zero $[x:t] \in \mathbb{P}^1$ with $t \neq 0$, the nonzero polynomial $S(x,1) \in \mathbb{C}[x]$ has no zero and is therefore constant. Homogeneity of degree N then gives $S = ct^N$, with $c \neq 0$, allowing $N = 0$. In the chart $y = 1$, take x, X from (5.15) and put $Y = 1$. The initial forms $\widehat{P},\widehat{Q}$ agree at these distinct points by the calculation following (5.15). The values of $\mathcal{R} = cy^{aN+r}$ also agree because $Y = y$. At these points, the derivative of $\mathcal{R}(x,y) - \mathcal{R}(X,Y)$ with respect to Y is $-c(aN+r) \neq 0$, and the Jacobian determinant of the differences for $\widehat{P},\widehat{Q}$ with respect to x, X is

$$36x^2(1-\omega^2) \neq 0 \quad (E), \qquad 48x \neq 0 \quad (H).$$

Thus the common zero is isolated.

In every case, $\mathcal{Z}(\widehat{P},\widehat{Q},\mathcal{R})$ has an isolated non-diagonal point. The positive integral weights $(a,1)$ allow Lemma 3.10 to produce collisions for $\boldsymbol{n}$, contradicting injectivity. □

6 Exclusion of $m_{\boldsymbol{n}} = 2$

Assume $m_{\boldsymbol{n}} = 2$ and choose the component Γ_i and coordinates from Proposition 3.6. Then

$$\Gamma_i = \{y = 0\}, \qquad a_i = 2, \qquad v = E(x^2 + c_0(y)), \qquad \mathrm{ord}_y\, c_0 \ge 2A_2.$$

We first determine the quadratic terms of $\boldsymbol{n}$ in Proposition 6.8. These terms will allow an analytic change of coordinates that preserves $\Gamma_i = \{y = 0\}$ and puts v in the form stated in Lemma 6.9. Divisibility of the Jacobian minors of $\boldsymbol{n}$ then reduces this map to the form in Proposition 6.23, which Proposition 6.27 excludes. We apply Steps 2 and 3 successively to refine the coordinates. In the final form of Proposition 6.23, the third coordinate must take different values at distinct points

in each fiber of the finite map (P, Q) formed by the first two coordinates. We find a pair violating this condition by studying a local component of the fiber product, with a second choice of weights when necessary. Put

$$A_j = \#\{p \in \mathcal{C} : m_{\boldsymbol{n},p} = j\} \quad (j = 1, 2), \qquad \mu_n = \{\zeta \in \mathbb{C}^* : \zeta^n = 1\} \quad (n \geq 1).$$

For every $p \in \mathcal{C}$, the multiplicity $m_{\boldsymbol{n},p}$ is one or two, and the origin is counted by A_2. Equations (2.1) and (2.5) become

$$m_X = 4A_2 + A_1, \qquad M_{\boldsymbol{n}} = 2A_2 + A_1, \qquad \delta(V, 0) = A_2, \tag{6.1}$$

$$\sum_i a_i(m_i - 1) \leq 6A_2 + A_1 - b, \tag{6.2}$$

where $b \in \{1, 2\}$.

6.1 The quadratic jet

Let

$$\mathcal{J}_2 = \operatorname{span}_{\mathbb{C}}\{(n_1)_2, (n_2)_2, (n_3)_2\},$$

where $(F)_2$ is the *homogeneous quadratic part* of F. Since $(d\boldsymbol{n})_0 = 0$, the triple $((n_1)_2, (n_2)_2, (n_3)_2)$ is the *quadratic jet* of $\boldsymbol{n}$. To show that $\mathcal{J}_2$ is generated by the square of a linear form, we first prove the following two lemmas. The first will also exclude the second numerical alternative in Proposition 6.16. Put

$$\Phi(s, \lambda) = s^2 - \lambda^2, \qquad \Psi(s, \lambda) = 2s^3 - 3s\lambda^2. \tag{6.3}$$

Lemma 6.4. *If $R \in \mathbb{C}[s, \lambda]$ is homogeneous of degree $r \geq 4$ and $R \notin \mathbb{C}[\Phi, \Psi]$, then $\mathcal{Z}(\Phi, \Psi, R)$ has an isolated point outside the diagonal.*

Proof. The identities $s^3 - 3\Phi s + \Psi = 0$ and $\lambda^2 = s^2 - \Phi$ make the polynomial map $(\Phi, \Psi) : \mathbb{C}^2 \to \mathbb{C}^2$ finite. In $\mathbb{P}^3$, with coordinates $[s : \lambda : s' : \lambda']$, set $D_\pm = \{s' = s, \lambda' = \pm\lambda\}$. On the locus $\Phi(s, \lambda) = \Phi(s', \lambda')$, factoring $\Psi(s, \lambda) - \Psi(s', \lambda')$ gives

$$\mathcal{Z}(\Phi, \Psi) = D_+ \cup D_- \cup \Gamma,$$

as sets, where Γ is defined by the equations

$$(s')^2 + ss' - 2s^2 + 3\lambda^2 = 0, \qquad (\lambda')^2 = (s')^2 - s^2 + \lambda^2. \tag{6.5}$$

The normalization map of Γ is the following morphism from the smooth conic $\mathcal{N} = \{\rho^2 = 3(\alpha^2 + \beta^2)\} \subset \mathbb{P}^2$:

$$\mathcal{N} \longrightarrow \Gamma,$$
$$[\rho : \alpha : \beta] \longmapsto [-(\alpha^2 + 2\beta^2) : \rho\beta : 2\alpha^2 + \beta^2 : \rho\alpha].$$

Indeed, the four quadratic forms defining this morphism have no common zero on $\mathcal{N}$, and their values satisfy (6.5). The morphism has inverse $[s' - s : \lambda' : \lambda]$ away from $o = [1 : 0 : 1 : 0]$. The point o is the only point of Γ with $s' = s$, and its preimages are $p_\pm = [0 : \pm i : 1]$. In particular, $\Gamma \cap (D_+ \cup D_-) = \{o\}$.

For $d \geq 0$, write $\mathcal{O}_{\mathcal{N}}(d)$ for the restriction of the line bundle $\mathcal{O}_{\mathbb{P}^2}(d)$ to $\mathcal{N}$. A homogeneous form of degree d restricts to a section of this line bundle. The pullback $\mathcal{R}$ of $R(s, \lambda) - R(s', \lambda')$ under the normalization map $\mathcal{N} \to \Gamma$ is a nonzero section of $\mathcal{O}_{\mathcal{N}}(2r)$. If it vanished identically, R would agree on every ordered pair in a general fiber of (Φ, Ψ) having different s-coordinates. Such a fiber has three distinct s-coordinates and two points above each, so R would be constant on the whole

fiber. Constancy on general fibers and separability of $\mathbb{C}(\Phi,\Psi)\subset\mathbb{C}(s,\lambda)$ would give $R\in\mathbb{C}(\Phi,\Psi)$. Finiteness makes R integral over $\mathbb{C}[\Phi,\Psi]$, which is a polynomial ring and hence integrally closed. Thus $R\in\mathbb{C}[\Phi,\Psi]$, contrary to the hypothesis.

The zero divisor of a nonzero section records its zeros with their orders of vanishing. Suppose the zeros of $\mathcal{R}$ lie only at $p_\pm$. Their orders sum to $4r$ by Bézout's theorem for the conic and a form of degree $2r$ [CA00, Section 2.6, p. 54]. Interchanging (s,λ) and (s',λ') exchanges p_+ and p_-, so both orders are $2r$. Since the pullback of $s'-s$ is ρ^2, which has order two at both points, both $\mathcal{R}$ and the pullback of $(s'-s)^r$ are sections of $\mathcal{O}_{\mathcal{N}}(2r)$ with zero divisor $2r(p_++p_-)$. Their quotient is a nowhere vanishing regular function on the projective conic $\mathcal{N}$, hence a nonzero constant. Thus $\mathcal{R}=c(s'-s)^r$, with $c\neq 0$, where the right-hand side denotes the pullback to $\mathcal{N}$. Interchanging the two factors negates $\mathcal{R}$ and multiplies $(s'-s)^r$ by $(-1)^r$, so r is odd. In the fiber $(\Phi,\Psi)=(1/3,0)$, take one point above each of $s=-1,0,1$. Summing the identity $R(s,\lambda)-R(s',\lambda')=c(s'-s)^r$ for the three ordered pairs with s-coordinates $(-1,0)$, $(0,1)$, and $(1,-1)$ yields $0=c(1+1-2^r)$, a contradiction. Thus $\mathcal{R}$ has a zero away from $p_\pm$. Its image is an isolated zero of $R(s,\lambda)-R(s',\lambda')$ on Γ, disjoint from the other components of $\mathcal{Z}(\Phi,\Psi)$, and outside the diagonal D_+. □

The second lemma concerns plane sections. For the choice of representatives of deformations in a Milnor ball, see [Ste85, Definition 2.2].

Lemma 6.6. *Let $(W,0)$ be a reduced plane curve germ of multiplicity four with two branches. There is no one-parameter flat deformation of $(W,0)$ represented by plane curves W_s in a fixed Milnor ball such that W_0 is singular only at the origin, all singular points of W_s tend to the origin as $s\to 0$, and the general fiber W_s, for sufficiently small $s\neq 0$, has only unibranch singularities of multiplicity two and satisfies*

$$\delta(W,0)=1+\sum_{Q\in\mathrm{Sing}(W_s)}\delta(W_s,Q). \tag{6.7}$$

Proof. We use the spectrum Sp as defined in [Ste85, (2.1)], with spectral numbers of plane curve singularities in $(-1,1)$ counted with multiplicity. Let $F=0$ be a reduced equation for $(W,0)$ in local coordinates (x,y). Write $F=F_4+H$, where F_4 is homogeneous of degree four and $H\in(x,y)^5\mathbb{C}\{x,y\}$. Put $G_t=F_4+t(x^4+y^4)$. Its discriminant is a nonzero polynomial in t, with leading coefficient the nonzero discriminant of x^4+y^4. Thus G_t is squarefree for all sufficiently small $t\neq 0$. For each such t, G_t has an isolated critical point at the origin, while uH has order at least five for every $u\in\mathbb{C}$. Hence [GLS07, Corollary I.2.18] gives $\mu(G_t+uH)=\mu(G_t)=9$. Invariance of the spectrum under deformations with constant Milnor number [Ste85, Theorem 2.8] therefore identifies $\mathrm{Sp}(F+t(x^4+y^4))$ with $\mathrm{Sp}(G_t)$. The spectrum formula for the homogeneous polynomial G_t [Ste89, p. 165] gives

$$\mathrm{Sp}(F+t(x^4+y^4))=\left\{\frac{i+j}{4}-1:1\le i,j\le 3\right\}.$$

Lower semicontinuity of the smallest spectral number [Ste85, Theorem 2.11], applied to the deformation $F+t(x^4+y^4)$, yields $\min\mathrm{Sp}(W)\le -1/2$. Spectral symmetry [Ste89, p. 167] gives $-\min\mathrm{Sp}(W)\ge 1/2$ as another spectral number. These two distinct spectral numbers lie outside $(-1/2,1/2)$, so at most $\mu(W,0)-2$ spectral numbers lie in that interval.

By [GLS07, Theorem I.2.48], each germ (W_s,Q), with $Q\in\mathrm{Sing}(W_s)$, has an equation $z^2-w^{q_Q}=0$ in suitable local coordinates, with $q_Q\ge 3$ odd by irreducibility. Its spectrum is

$$\left\{-\frac{1}{2}+\frac{j}{q_Q}:1\le j<q_Q\right\}\subset(-1/2,1/2)\qquad\text{[Ste89, p. 165].}$$

Under the stated assumptions on the representatives W_s, spectral semicontinuity applies to the number of spectral numbers in each open interval of length one [Ste85, Remark 2.10]. Hence

$$\sum_Q \mu(W_s, Q) \le \#\big(\mathrm{Sp}(W) \cap (-1/2, 1/2)\big) \le \mu(W, 0) - 2.$$

Applying $\mu = 2\delta - b + 1$ to $(W, 0)$ and each (W_s, Q), with b denoting the number of branches of the germ in question, and using (6.7) instead gives $\sum_Q \mu(W_s, Q) = 2\delta(W, 0) - 2 = \mu(W, 0) - 1$, a contradiction. □

We will also use the *splitting lemma with one coordinate fixed*. For a holomorphic function germ $f \in \mathbb{C}\{x, y\}$ with $f_x(0) = 0$ and $f_{xx}(0) \neq 0$, the implicit function theorem gives a holomorphic germ $\xi(y)$ with $\xi(0) = 0$ and $f_x(\xi(y), y) = 0$. Taylor expansion gives

$$f(x, y) = f(\xi(y), y) + (x - \xi(y))^2 U(x, y), \qquad U(0) \neq 0.$$

Thus the change of coordinates $(x, y) \mapsto (X, y)$, where $X = (x - \xi(y))\sqrt{U(x, y)}$, puts f in the form $X^2 + h(y)$, where $h(y) = f(\xi(y), y)$, while preserving y. This proves the form of [GLS07, Theorem I.2.47] that preserves the parameter y, as needed below.

The preceding lemmas now determine $\mathcal{J}_2$.

Proposition 6.8. *There is a nonzero linear form ℓ such that $\mathcal{J}_2 = \mathrm{span}_{\mathbb{C}}\{\ell^2\}$.*

Proof. If $\dim \mathcal{J}_2 = 3$, choose target coordinates so that the quadratic initial forms of the components of $\boldsymbol{n}$ are x^2, xy, y^2. Their Jacobian minors $2x^2, 4xy, 2y^2$ have no common factor, contradicting the nontrivial common divisor in (2.7).

Suppose $\dim \mathcal{J}_2 = 2$ and all elements of $\mathcal{J}_2$ have a nonconstant common factor, so $\mathcal{J}_2 = x\,\mathrm{span}_{\mathbb{C}}\{x, y\}$ in suitable coordinates. By Proposition 3.6, the chosen component Γ_i is smooth and $i_0(v, \Gamma_i) = 2$. Let L be the linear initial form of a reduced defining function of Γ_i. Divisibility of the Jacobian minors gives

$$L^{m_{\mathrm{tr}}-1} \mid J(x^2, xy) = 2x^2.$$

Thus L is proportional to x. Since the quadratic part of v is also divisible by x, the restriction of v to Γ_i has order greater than two, a contradiction.

Suppose instead that the two generators of $\mathcal{J}_2$ have no common zero in $\mathbb{P}^1$. The ideal I has no base point after the blowup at the origin, so $A_2 = 1, A_1 = 0, m_X = 4, b = 2$. Then $M_{\boldsymbol{n}} = 2$, and Proposition 3.6 gives $m_{\mathrm{tr}} = 2$. Hence every $m_i = 2$. Write q_v, q_u for the coprime quadratic initial forms of v, u.

The two branches of V are smooth. On each, u has order two because q_u does not vanish at either root of q_v. The images of the two branches of V under $\boldsymbol{n}$ are the branches of W and therefore have multiplicity two, so W has multiplicity four and two branches.

The sections $W_s = X \cap \{\lambda_1 = s\}$ form a one-parameter flat deformation of $(W, 0)$ because the local ring of the irreducible surface X at the origin is torsion-free over the discrete valuation ring $\mathbb{C}\{\lambda_1\}$ [GLS07, Theorem B.8.11]. By [PC26, Proposition 2.10 and Section 2.2], the germs (W_s, Q) at the nearby singular points are the transverse branches of X along Σ, and

$$\delta(W, 0) = \delta(V, 0) + \sum_{Q \in \mathrm{Sing}(W_s)} \delta(W_s, Q).$$

Choose representatives of W_s transverse to the boundary of a Milnor ball for $(W, 0)$. All nearby singular points tend to the origin since $\Sigma \cap \{\lambda_1 = 0\} = \{0\}$. Since $\delta(V, 0) = 1$ and every $m_i = 2$, this contradicts Lemma 6.6.

It remains to exclude a one-dimensional $\mathcal{J}_2$ with a squarefree generator. The two simple zeros of this generator in $\mathbb{P}^1$ yield base points p with $m_{\boldsymbol{n},p} = 1$, so $A_2 = 1, A_1 \geq 2, b = 2$. For the chosen component Γ_i of the critical locus with $m_i = m_{\mathrm{tr}}$, (2.4) and (6.2) give

$$2(A_1 + 1) \leq a_i(m_{\mathrm{tr}} - 1) \leq A_1 + 4.$$

The inequalities $2(A_1 + 1) \leq A_1 + 4$ and $A_1 \geq 2$ give $A_1 = 2$. Proposition 3.6 gives $a_i = 2$ and a smooth Γ_i. Since $m_{\mathrm{tr}} \geq M_{\boldsymbol{n}} = 4$, the displayed inequalities force $m_{\mathrm{tr}} = 4$. Take a reduced defining function of Γ_i as a coordinate λ. The splitting lemma with λ fixed, followed by multiplication of λ by a unit, gives $v = s^2 - \lambda^2 = \Phi$, since the restriction of v to $\lambda = 0$ has order two and the Hessian of v is nondegenerate.

After subtracting scalar multiples of v, the other two coordinate functions of $\boldsymbol{n}$ have order at least three. Both branches of W then have multiplicity at least three. If both remaining coordinates had order at least four, each image branch would have multiplicity at least four, contrary to $\mathrm{mult}_0 W = m_X = 6$. Thus some remaining coordinate has a nonzero cubic part H_3. The divisibility $\lambda^3 \mid J_{s,\lambda}(\Phi, H_3)$ forces $H_3 \in \mathrm{span}_{\mathbb{C}}\{2s^3 - 3s\lambda^2\}$, by comparing the coefficients of $\lambda^0, \lambda^1, \lambda^2$ in $J_{s,\lambda}(\Phi, H_3)$. After a linear change of target coordinates, the homogeneous parts of least degree in the first two coordinate functions of $\boldsymbol{n}$ are Φ, Ψ, and the third coordinate has order at least four. The initial forms Φ, Ψ have degrees two and three and no common zero in $\mathbb{P}^1$, so their intersection multiplicity is six. By injectivity and Lemma 5.16, subtracting a polynomial in the first two coordinate functions from the third gives a third coordinate whose initial form $R \notin \mathbb{C}[\Phi, \Psi]$ has degree at least four. Lemmas 3.10 and 6.4 contradict injectivity. Thus $\mathcal{J}_2$ is spanned by the square of a linear form. □

6.2 Divisibility of the Jacobian minors and the numerical alternatives

We now refine the coordinates of Proposition 3.6 while keeping Γ_i as the axis $y = 0$. Since $\mathcal{J}_2$ is generated by the square of a linear form, the splitting lemma applies to v with y fixed. Divisibility of the Jacobian minors along Γ_i will then give a recurrence for the coefficients of the coordinate functions of $\boldsymbol{n}$. Put $k = 2A_2 - b + 2$.

Lemma 6.9. *The integer k satisfies $k \geq 3$. There are local coordinates x, y such that*

$$v = x^2 + y^k, \qquad I(\Gamma_i) = (y).$$

Here $I(\Gamma_i)$ is the ideal of functions in B vanishing on Γ_i.

Proof. Return to the coordinates of Proposition 3.6. Since $v(x, 0)$ has order two, $v_{xx}(0) \neq 0$. The splitting lemma with y fixed gives $v = x^2 + h(y)$ and preserves $\Gamma_i = \{y = 0\}$. By Proposition 6.8, the Hessian of v has rank one, so $\mathrm{ord}_y h \geq 3$. Reducedness of v gives $h \neq 0$, since $h = 0$ would give $v = x^2$. The Jacobian ideal of v is $(x, h'(y)) \subset \mathbb{C}\{x, y\}$, so $\mu(V, 0) = \dim_{\mathbb{C}} \mathbb{C}\{y\}/(h'(y)) = \mathrm{ord}_y h - 1$, and (6.1) and (2.6) yield

$$\mathrm{ord}_y h = \mu(V, 0) + 1 = 2A_2 - b + 2 = k.$$

Since $h(y)/y^k$ is a unit, replacing y by $y(h(y)/y^k)^{1/k}$ gives the required form and preserves $\Gamma_i = \{y = 0\}$. □

Fix the coordinates and component of Lemma 6.9, so $a_i = 2$. The branch formula (3.3) gives $A_1 \geq 2 - b$. The inequalities $m_{\mathrm{tr}} \geq M_{\boldsymbol{n}} = 2A_2 + A_1$ and $2(m_{\mathrm{tr}} - 1) \leq 6A_2 + A_1 - b$ give $A_1 \leq k$. Substituting $2A_2 = k + b - 2$ yields

$$2 - b \leq A_1 \leq k, \qquad k \leq m_{\mathrm{tr}} \leq \frac{3k + 2b - 4 + A_1}{2} \leq 2k. \tag{6.10}$$

Every $F \in B$ has a unique decomposition

$$F = A_F(v,y) + xB_F(v,y), \qquad B_F(v,y) = \sum_{j\geq 0} b_j(v)y^j,$$

with $A_F, B_F \in \mathbb{C}\{v,y\}$. The summands $A_F(v,y)$ and $xB_F(v,y)$ are respectively its *even part* and *odd part* with respect to $x \mapsto -x$. Indeed, in the power series expansion of F, group the even powers of x and factor one x from the odd powers. The two remaining series lie in $\mathbb{C}\{x^2,y\} = \mathbb{C}\{v,y\}$, since $x^2 = v - y^k$. Uniqueness follows because the even and odd powers of x cannot cancel. Thus $B = \mathbb{C}\{v,y\} \oplus x\mathbb{C}\{v,y\}$ as a module over $\mathbb{C}\{v,y\}$. The points (x,y) and $(-x,y)$ have the same values of v,y. The even part takes the same value at these points, and the difference of their F-values is $2xB_F(v,y)$. For pullbacks of target coordinates, the Jacobian divisibility in (2.7) constrains the odd part of F, and (6.10) leaves at most two nonzero coefficients of B_F below order m_{tr} in y. In derivatives such as $(A_F)_y$ and $(B_F)_y$, the variable v is held fixed.

Lemma 6.11. *Let F be the pullback under $\boldsymbol{n}$ of a target coordinate after a biholomorphic change of target coordinates preserving v. There are $\alpha_F, \beta_F \in \mathbb{C}\{v\}$, with $\alpha_F(0) = 0$, such that*

$$F \equiv \begin{cases} \alpha_F(v) + x\beta_F(v), & m_{\mathrm{tr}} = k, \\ \alpha_F(v) + x(2v + y^k)\beta_F(v), & k < m_{\mathrm{tr}} \leq 2k, \end{cases} \pmod{y^{m_{\mathrm{tr}}}}. \tag{6.12}$$

In the first case, $\beta_F(0) = 0$.

Proof. Changes of target coordinates preserve (2.7), since the transformed minors are analytic linear combinations of the original minors. Hence

$$J(v,F) = 2x(A_F)_y + 2(v - y^k)(B_F)_y - ky^{k-1}B_F \in (y^{m_{\mathrm{tr}}-1}).$$

Separating even and odd powers of x gives $(A_F)_y \in (y^{m_{\mathrm{tr}}-1})$ and

$$2vjb_j = (2j-k)b_{j-k} \quad (1 \leq j < m_{\mathrm{tr}}), \qquad b_j = 0 \quad (j < 0).$$

By (6.10), $k \leq m_{\mathrm{tr}} \leq 2k$, so all coefficients b_j with $0 \leq j < m_{\mathrm{tr}}$ vanish except possibly b_0 and, when $m_{\mathrm{tr}} > k$, b_k. In the latter case the recurrence at $j = k$ is $b_0 = 2vb_k$, giving the second congruence. Integrating $(A_F)_y \in (y^{m_{\mathrm{tr}}-1})$ with respect to y gives $A_F \equiv \alpha_F(v) \pmod{y^{m_{\mathrm{tr}}}}$. Finally, $F(0) = 0$ gives $\alpha_F(0) = 0$, and in the first case $(dF)_0 = 0$ also gives $\beta_F(0) = 0$. □

Lemma 6.11 bounds the orders of pullbacks of target coordinates along the branches of V. For a branch C of V with normalization parametrization $\gamma_C : (\mathbb{C},0) \to (C,0)$, write $\nu_C(F) = \mathrm{ord}_t(F \circ \gamma_C)$. We set $\nu_C(F) = \infty$ when F vanishes identically on C.

Corollary 6.13. *For every F as in Lemma 6.11 and every branch C of V,*

$$b\nu_C(F) \geq \min\{2m_{\mathrm{tr}}, 3k\}. \tag{6.14}$$

Proof. On the normalization of each branch C of V, $\nu_C(y) = 2/b$ and $\nu_C(x) = k/b$. If $m_{\mathrm{tr}} = k$, the first congruence in (6.12) shows that F restricts to zero modulo y^k on V, so $b\nu_C(F) \geq 2k$. Otherwise, the second congruence shows that F restricts to $\beta_F(0)xy^k$ modulo $y^{m_{\mathrm{tr}}}$ on V. The monomial xy^k has order $3k/b$, and the remainder has order at least $2m_{\mathrm{tr}}/b$. □

We compare the orders along the branches in (6.14) with $i_0(v,u) = m_X$, from (2.2).

Proposition 6.15. *The integer k is even, and*

$$(A_1, m_{\mathrm{tr}}) = (0,k) \qquad \text{or} \qquad (A_1, m_{\mathrm{tr}}) = (k, 2k).$$

Proof. Summing (6.14) for $F = u$ over the branches of V gives

$$m_X \geq \min\{2m_{\mathrm{tr}}, 3k\}.$$

On the other hand, (6.1) and (6.10) give

$$2m_{\mathrm{tr}} \geq 2M_{\boldsymbol{n}} = m_X + A_1, \qquad m_X = 2k + 2b - 4 + A_1 \leq 3k.$$

If $A_1 > 0$, then $2m_{\mathrm{tr}} > m_X$, so the lower bound forces $m_X = 3k$. Since $b \leq 2$ and $A_1 \leq k$, the equality $m_X = 3k$ forces $b = 2$ and $A_1 = k$. Now $2k = M_{\boldsymbol{n}} \leq m_{\mathrm{tr}} \leq 2k$.

If $A_1 = 0$, the bound $A_1 \geq 2 - b$ gives $b = 2$, hence $m_X = 2k$. The lower bound for m_X, together with $m_{\mathrm{tr}} \geq k$, then forces $m_{\mathrm{tr}} = k$. In both cases $b = 2$, so $k = 2A_2$ is even. □

6.3 Exclusion of the second alternative

For $(A_1, m_{\mathrm{tr}}) = (k, 2k)$, Lemma 6.11 determines the initial forms of the first two coordinate functions. These forms are obtained from (Φ, Ψ) in (6.3) by setting $s = x$ and $\lambda = iy^{k/2}$. We first compare the initial form of the third coordinate at (x, y) and $(x, \zeta y)$, where $\zeta^k = 1$, and then use Lemma 6.4.

Proposition 6.16. *The alternative $(A_1, m_{\mathrm{tr}}) = (k, 2k)$ is impossible.*

Proof. Write $k = 2a$ and give x, y weights $a, 1$. The recurrence in Lemma 6.11 gives, for the pullback F of every target coordinate,

$$F \equiv \alpha_F(v) + \beta_F(v)\Theta \pmod{y^{2k}}, \qquad \Theta = x(2v + y^k).$$

The coefficient $\beta_u(0)$ is nonzero, since otherwise u would have order at least $2k$ on each of the two branches of V. This would give $i_0(v, u) \geq 4k$, contrary to $i_0(v, u) = m_X = 3k$. Set $U = (u - \alpha_u(v))/\beta_u(v)$ and replace the third coordinate w by $Z = w - \alpha_w(v) - \beta_w(v)U$. The map then has the form

$$\boldsymbol{n} = (G, U, Z) = (v, \Theta + y^{2k}C_2, y^{2k}C_3), \qquad C_2, C_3 \in B.$$

The initial forms of G and U for weights $(a, 1)$ are, respectively,

$$v_0 = x^2 + y^k, \qquad \Theta_0 = 2x^3 + 3xy^k.$$

They have only the origin as a common zero, and $i_0(G, U) = m_X = 3k > 1$. By injectivity and Lemma 5.16, subtracting a polynomial in G, U from the third coordinate gives a third coordinate whose initial form $R \notin \mathbb{C}[v_0, \Theta_0]$ has weight $\rho \geq 2k = 4a$.

Write $\mu_k = \{\zeta \in \mathbb{C} : \zeta^k = 1\}$ for the group of kth roots of unity. Use (x, y) and (X, Y) for the two source points in the fiber product of (v_0, Θ_0). For $\zeta \in \mu_k$, the graph $X = x, Y = \zeta y$ lies in this fiber product. Since $J(v_0, \Theta_0) = -3ky^{2k-1}$, the graph is the unique local component of this fiber product at each of its points with $y \neq 0$. Consequently, if the polynomial

$$R(x, \zeta) - R(x, 1)$$

is nonzero, each of its zeros gives an isolated collision of (v_0, Θ_0, R) in the chart $y = 1$. By Lemma 3.10, injectivity forces this polynomial to be constant for every ζ. Comparing the coefficients of positive powers of x shows that the exponent of y in each monomial of R involving x is divisible by k. Hence

$$R = U_0(x, y^k) + cy^\rho, \tag{6.17}$$

where $U_0 \in \mathbb{C}[x,t]$ is homogeneous for weights a, k on x, t, respectively.

If $a \mid \rho$, substituting $s = x, \lambda = iy^a$ gives $v_0 = \Phi(s,\lambda)$, $\Theta_0 = \Psi(s,\lambda)$, and $R = \widehat{R}(s,\lambda)$, where $\widehat{R}$ is homogeneous of degree $\rho/a \geq 4$. Since $\widehat{R} \notin \mathbb{C}[\Phi,\Psi]$, Lemma 6.4 gives an isolated collision of $(\Phi,\Psi,\widehat{R})$. The map

$$\mathbb{P}(a,1,a,1) \longrightarrow \mathbb{P}^3, \qquad [x:y:X:Y] \longmapsto [x:iy^a:X:iY^a]$$

is well defined since its four coordinate polynomials have weighted degree a. It is finite because the corresponding extension of homogeneous coordinate rings adjoins a-th roots of the second and fourth target coordinates divided by i, and surjective because these roots exist over $\mathbb{C}$. The difference equations for (v_0,Θ_0,R) are pullbacks of those for $(\Phi,\Psi,\widehat{R})$, so $\mathcal{Z}(v_0,\Theta_0,R)$ is the inverse image of $\mathcal{Z}(\Phi,\Psi,\widehat{R})$. The inverse image of the isolated collision is nonempty and consists of isolated points outside the diagonal, so Lemma 3.10 contradicts injectivity.

If $a \nmid \rho$, homogeneity in (6.17) forces $R = cy^\rho$, with $c \neq 0$. Choose $\xi^2 = -3/2$. The points $(\xi,1)$ and $(-\xi,1)$ have the same image under (v_0,Θ_0,R). In the chart $y = 1$, take the differences between the values of v_0, Θ_0, R at $(x,1)$ and (X,Y). The Jacobian matrix of these differences with respect to x, X, Y, evaluated at $(x,X,Y) = (\xi,-\xi,1)$, is

$$\begin{pmatrix} 2\xi & 2\xi & -k \\ -6 & 6 & 3k\xi \\ 0 & 0 & -c\rho \end{pmatrix},$$

with determinant $-24c\rho\xi \neq 0$. The collision is isolated, and Lemma 3.10 again contradicts injectivity. □

6.4 The form of the map along $y = 0$

In the remaining alternative $(A_1, m_{\mathrm{tr}}) = (0,k)$, the congruence (6.12) gives $F(x,y) \equiv F(x,0) \pmod{y^k}$ for every coordinate function F of $\boldsymbol{n}$. We change target coordinates to simplify the restriction of $\boldsymbol{n}$ to $y = 0$ and obtain the form used in the final collision argument.

Proposition 6.18. *There are an odd integer $p \geq 3$ and source and target coordinates in which*

$$\boldsymbol{n}(x,y) = \bigl(x^2 + y^k, x^p + y^k h_2(x,y), y^k h_3(x,y)\bigr), \tag{6.19}$$

where $k \geq 4$ is even and $h_2, h_3 \in \mathbb{C}\{x,y\}$ satisfy

$$(h_2(0), h_3(0)) \neq (0,0).$$

Proof. Propositions 6.15 and 6.16 give $m_{\mathrm{tr}} = k \geq 4$ and $m_X = 2k$. Lemma 6.11 gives $F(x,y) \equiv \alpha_F(v) + x\beta_F(v) \pmod{y^k}$. Since $v \equiv x^2 \pmod{y^k}$, it follows that $F(x,y) \equiv F(x,0) \pmod{y^k}$.

The restriction $\gamma = \boldsymbol{n}|_{y=0}$ is injective and has first coordinate x^2. Subtract from the other two coordinates of $\boldsymbol{n}$ their even parts on $y = 0$, expressed as functions of the first coordinate v. At least one of the other two coordinate functions restricts to a nonzero odd function on $y = 0$, since γ is injective. Choose as the second coordinate one whose restriction to $y = 0$ has least order $p \geq 3$, necessarily odd. This restriction is $x^p\eta(x^2)$, with $\eta(0) \neq 0$. Dividing the second coordinate by η evaluated at the first coordinate makes its restriction equal to x^p. By minimality of p, the restriction of the third coordinate is $x^p\xi(x^2)$ for a convergent series ξ. Subtracting $\xi(v)$ times the second coordinate from the third gives $\gamma = (x^2, x^p, 0)$. The congruences modulo y^k are preserved under these target changes. The transformed coordinates therefore satisfy (6.19).

If $h_2(0) = h_3(0) = 0$, the last two coordinate functions of $\boldsymbol{n}$ in (6.19) would have order greater than k on each branch of V, since $\nu_C(x^p) = kp/2 > k$ and $\nu_C(y^k h_j) > k$. The original u, an analytic function of the new coordinates vanishing at the origin, would also have order greater than k on each branch of V, because the first coordinate vanishes on V. Then $i_0(v,u) > 2k = m_X$, contrary to (2.2). □

6.5 The coordinate $Q = x^p + y^k\beta(y)$ and the weighted initial form of Z

By Proposition 6.18,

$$\boldsymbol{n} = (P, Q_0, Z_0) = \big(x^2 + y^k, x^p + y^k h_2, y^k h_3\big), \qquad (h_2(0), h_3(0)) \neq (0,0), \tag{6.20}$$

where $k \geq 4$ is even, $p \geq 3$ is odd, and $h_2, h_3 \in \mathbb{C}\{x,y\}$. To study the fiber product of the first two coordinates in Step 3, we seek coordinates in which

$$(P, Q, Z) = \big(x^2 + y^k, x^p + y^k\beta(y), Z(x,y)\big), \qquad \beta \in \mathbb{C}\{y\}, \quad Z \in \mathbb{C}\{x,y\}, \quad \beta(0) = c \neq 0, \tag{6.21}$$

with

$$\operatorname{in}_{(k,p)} Z = x^N, \qquad N \geq 3 \text{ odd}. \tag{6.22}$$

In these coordinates, the third coordinate Z must distinguish the points in each fiber of the finite map (P, Q). Thus any distinct points with equal P, Q, Z values contradict injectivity, as in (1.2).

Proposition 6.23. *Every injective germ in* (6.20) *can be transformed by biholomorphic changes of source and target coordinates into a germ satisfying* (6.21) *and* (6.22).

Proof. Putting Q in the form $x^p + y^k\beta(y)$. For a suitable $\alpha \in \mathbb{C}$, the linear change $Q_0 \mapsto Q_0 + \alpha Z_0$ gives $Q = x^p + y^k h(x,y)$ with $h(0) = c \neq 0$. For an unknown convergent series $\theta \in \mathbb{C}\{x,y\}$, set

$$A_\theta = 1 - 2x\theta - y^k\theta^2, \qquad X_\theta = x + y^k\theta, \qquad Y_\theta = yA_\theta^{1/k},$$

using the kth root of A_θ with constant term one. This substitution preserves P and the ideal (y^k), and fixes the axis $y = 0$ pointwise. Moreover,

$$Q(X_\theta, Y_\theta) = x^p + y^k\mathcal{H}(\theta), \qquad \mathcal{H}(\theta) = \frac{(x + y^k\theta)^p - x^p}{y^k} + A_\theta h(X_\theta, Y_\theta).$$

The quotient in this expression is analytic. We seek an analytic function $\theta(x,y)$ for which $\mathcal{H}(\theta(x,y))$ depends only on y. Regard the argument of $\mathcal{H}$ temporarily as an independent variable t. At $y = 0$,

$$\mathcal{H}(x,0,t) = h(x,0) + t\big(px^{p-1} - 2xh(x,0)\big).$$

Put $t_0 = h_x(0)/(2c)$. The point $(x,t) = (0,t_0)$ is critical for $\mathcal{H}(x,0,t)$, and its Hessian in (x,t) has determinant $-4c^2 \neq 0$. The implicit-function theorem gives an analytic critical point $(a(y), b(y))$ of $(x,t) \mapsto \mathcal{H}(x,y,t)$ with $(a(0), b(0)) = (0,t_0)$. Let $\beta(y) = \mathcal{H}(a(y), y, b(y))$ be the corresponding critical value, so $\beta(0) = c$. For each fixed y near zero, the level curve $\mathcal{H}(x,y,t) = \beta(y)$ has two transverse analytic branches at $(a(y), b(y))$. We choose a branch of the form $t = \theta(x,y)$ by the following factorization.

Write $\xi = x - a(y)$ and $\tau = t - b(y)$. Taylor expansion at the critical point gives

$$\mathcal{H} - \beta = a_{20}\xi^2 + 2a_{11}\xi\tau + a_{02}\tau^2, \qquad a_{11}(0) = -c, \quad a_{02}(0) = 0,$$

where a_{20}, a_{11}, a_{02} are analytic in (ξ, τ, y). Choose the analytic square root S of $a_{11}^2 - a_{20}a_{02}$ with $S(0) = -c$. Since $a_{11} + S$ is a unit,

$$\mathcal{H} - \beta = (a_{20}\xi + (a_{11} + S)\tau)\left(\xi + \frac{a_{02}}{a_{11} + S}\tau\right).$$

The first factor has nonzero t-derivative at $(x,y,t) = (0,0,t_0)$. The implicit-function theorem therefore gives an analytic $t = \theta(x,y)$ on which $\mathcal{H} = \beta(y)$. The derivative of (X_θ, Y_θ) at the origin

is the identity. This change of coordinates preserves the function P and the ideal (y^k), and fixes $y = 0$ pointwise.

After this substitution, the second coordinate is $x^p + y^k\beta(y)$, which we again denote by Q. The transformed third coordinate is $\widetilde{Z}_0 = y^k\widetilde{h}_3$, with $\widetilde{h}_3 \in \mathbb{C}\{x, y\}$. Writing $p = 2m + 1$, we have

$$Q \equiv y^k(\beta(y) + (-1)^m xy^{k(m-1)}) \pmod{P}, \qquad (P, Q)\mathbb{C}\{x, y\} = (x^2, y^k). \tag{6.24}$$

The factor $\beta(y) + (-1)^m xy^{k(m-1)}$ is a unit because its constant term is $c \neq 0$. The quotient ring $\mathbb{C}\{x, y\}/(x^2, y^k)$ has $\mathbb{C}$-basis $1, y, \ldots, y^{k-1}, x, xy, \ldots, xy^{k-1}$. Thus (P, Q) is finite of degree $2k$, by Lemma 5.16, and injectivity implies $\widetilde{Z}_0 \neq 0$. If $d = \widetilde{h}_3(0) \neq 0$, replacing the third coordinate by $Q - (c/d)\widetilde{Z}_0$ gives initial form x^p for weights (k, p), as required. For the remaining case, write the transformed third coordinate as $Z_0 = y^k h$, with $0 \neq h \in (x, y)$.

Subtracting functions of P, Q. Use weights (k, p) on (x, y), and put

$$P_{\text{in}} = x^2, \quad Q_{\text{in}} = x^p + cy^k, \quad A = \mathbb{C}\{P, Q\}, \quad \mathcal{R} = Q^2 - P^p.$$

The polynomial map $(P_{\text{in}}, Q_{\text{in}})$ is finite, so Lemma 5.16 applies and $\operatorname{gr} A = \mathbb{C}[P_{\text{in}}, Q_{\text{in}}]$. Identify $\mathbb{C}[S, T]$ with $\mathbb{C}[P_{\text{in}}, Q_{\text{in}}]$ by $S \mapsto P_{\text{in}}$, $T \mapsto Q_{\text{in}}$. Restriction to $y = 0$ is the homomorphism

$$\mathbb{C}[S, T] \longrightarrow \mathbb{C}[x], \qquad S \longmapsto x^2, \quad T \longmapsto x^p.$$

Division of a polynomial in $\mathbb{C}[S, T]$ by $T^2 - S^p$, monic in T, leaves a remainder $A_0(S) + TA_1(S)$. Its image is $A_0(x^2) + x^pA_1(x^2)$. Since p is odd, the first summand has only even powers of x and the second only odd powers. Their sum vanishes only when $A_0 = A_1 = 0$. The kernel of the restriction homomorphism is therefore $(Q_{\text{in}}^2 - P_{\text{in}}^p)$ in $\mathbb{C}[P_{\text{in}}, Q_{\text{in}}]$. Expansion of $Q^2 - P^p$ gives $\mathcal{R} \in y^k(x, y)$ and $\operatorname{in}_{(k,p)} \mathcal{R} = Q_{\text{in}}^2 - P_{\text{in}}^p$. Starting with Z_0, whenever the current remainder has initial form in $\operatorname{gr} A$, we cancel that initial form by subtracting $\mathcal{R}C(P, Q)$ for a polynomial $C(S, T)$ weighted homogeneous with $\deg S = 2k$ and $\deg T = pk$. These subtractions preserve $y^k(x, y)$. The proof of Lemma 5.16 shows that they terminate unless $Z_0 \in A$, which injectivity excludes by (6.24). The remainder obtained from Z_0 therefore has initial form

$$H = y^k\Lambda \notin \mathbb{C}[P_{\text{in}}, Q_{\text{in}}], \tag{6.25}$$

where Λ is weighted homogeneous of positive degree.

In this proof, a polynomial is invariant if it is unchanged by every substitution $(x, y) \mapsto (x, \zeta y)$ with $\zeta \in \mu_k$. A monomial x^iy^j is invariant under this action exactly when $k \mid j$. For $\zeta \in \mu_k \setminus \{1\}$, let $D_\zeta(s) = \Lambda(s, 1) - \Lambda(s, \zeta)$. A nonzero root s_0 of a nonzero D_ζ would give an isolated collision of $(P_{\text{in}}, Q_{\text{in}}, H)$ at $(s_0, 1), (s_0, \zeta)$, since locally the first two difference equations select $X = x$, $Y = \zeta y$, and the difference of the H values restricts to D_ζ after setting $y = 1$. Consequently, injectivity and Lemma 3.10 imply that a nonzero $D_\zeta \in \mathbb{C}[s]$ has no root outside $s = 0$. Factoring this polynomial over $\mathbb{C}$ shows that it is a nonzero scalar multiple of a power of s, allowing exponent zero. Thus every D_ζ is either identically zero or a monomial. When ζ is a primitive kth root of unity, distinct noninvariant monomials of Λ give distinct powers of s, by weighted homogeneity. Hence either $\Lambda \in \mathbb{C}[x, y^k]$, or

$$\Lambda = \Lambda_{\text{inv}} + \gamma x^i y^j, \quad \Lambda_{\text{inv}} \in \mathbb{C}[x, y^k], \quad \gamma \neq 0, \quad i, j \geq 0, \quad k \nmid j. \tag{6.26}$$

We exclude (6.26) by considering pairs $(1, y), (-1, Y)$ with equal values of $P_{\text{in}}, Q_{\text{in}}$. These equalities define the affine curve

$$C = \{(y, Y) \in \mathbb{C}^2 : Y^k - y^k = 2/c\}.$$

Using $Y^k = y^k + 2/c$, write $y^k\Lambda_{\mathrm{inv}}(1,y) - Y^k\Lambda_{\mathrm{inv}}(-1,Y)$ as $B(y^k)$ on C. Let $\mathcal{G} \in \mathbb{C}[y,Y]$ be the polynomial on the right below. On C,

$$\mathcal{G}(y,Y) = H(1,y) - H(-1,Y) = B(y^k) + \gamma(y^{k+j} - (-1)^i Y^{k+j}), \qquad B \in \mathbb{C}[z].$$

To obtain the collision required in Step 3, we show that $\mathcal{G}$ has an isolated zero on C. We prove that its homogenization has a zero on the projective closure of C away from infinity.

The projective closure of C,

$$\overline{C} = \{[y:Y:T] \in \mathbb{P}^2 : Y^k - y^k = (2/c)T^k\},$$

is smooth because the three partial derivatives of its defining polynomial have no common projective zero. It is irreducible because distinct projective plane components would meet and produce a singular point. Let $\mathcal{G}_{\mathrm{hom}}$ be the homogenization of $\mathcal{G}$ with respect to T. The line at infinity is $T = 0$. If $B \neq 0$ and $k \deg B > k + j$, then $\mathcal{G}_{\mathrm{hom}}$ equals the nonzero leading coefficient of B at every point $[1:\zeta:0]$ of $\overline{C}$ at infinity. Bézout's theorem [CA00, Section 2.6, p. 54] then gives an affine zero of $\mathcal{G}|_C$, since $\overline{C}$ and $V(\mathcal{G}_{\mathrm{hom}})$ have positive degrees and no intersection at infinity.

Otherwise, $k \nmid j$ implies that $\mathcal{G}$ has total degree $k + j$. At infinity its homogenization has values $\gamma(1 - (-1)^i\zeta^{k+j})$, where $\zeta^k = 1$. These values are not all zero, since $k \nmid j$. Thus $\mathcal{G}_{\mathrm{hom}}$ does not vanish identically on $\overline{C}$, and the total intersection number of $V(\mathcal{G}_{\mathrm{hom}})$ with $\overline{C}$ is $k(k+j)$. At a zero of $\mathcal{G}_{\mathrm{hom}}$ on $\overline{C}$ at infinity, use T as local parameter on $\overline{C}$ in the chart $y = 1$:

$$Y = \zeta(1 + 2T^k/c)^{1/k}, \qquad \zeta^k = 1, \qquad (-1)^i\zeta^{k+j} = 1.$$

The contribution of $\gamma(y^{k+j} - (-1)^i Y^{k+j})$ to $\mathcal{G}_{\mathrm{hom}}$ in this parametrization is

$$\gamma(1 - (1 + 2T^k/c)^{(k+j)/k}),$$

which has order exactly k. The contribution of $B(y^k)$ has T-exponents congruent to j modulo k, so it cannot cancel the coefficient of T^k. The intersection multiplicity at each of the k points at infinity is therefore at most k. Since $j > 0$, the sum of these multiplicities satisfies

$$\sum_{a \in \overline{C} \cap \{T=0\}} i_a(\overline{C}, V(\mathcal{G}_{\mathrm{hom}})) \leq k^2 < k(k+j).$$

Hence an affine zero of $\mathcal{G}|_C$ exists in this case as well.

In both cases, the restriction of $\mathcal{G}$ to the irreducible curve C does not vanish identically, so its zeros are isolated. Any such zero gives distinct points $(1,y), (-1,Y)$ with equal values of $P_{\mathrm{in}}, Q_{\mathrm{in}}, H$. Setting $x = 1$ gives a finite cover of the weighted projective chart $x \neq 0$. On this cover, the first difference equation locally selects $X = -1$, and the remaining equations are those of $C \cap V(\mathcal{G})$. The collision is therefore isolated in $\mathcal{Z}(P_{\mathrm{in}}, Q_{\mathrm{in}}, H)$. By Lemma 3.10, this contradicts injectivity. Therefore $\Lambda \in \mathbb{C}[x, y^k]$.

The case $\Lambda \in \mathbb{C}[x, y^k]$. Set $t = y^k$ and give x, t degrees $1, p$. Since $t = c^{-1}(Q_{\mathrm{in}} - xP_{\mathrm{in}}^{(p-1)/2})$,

$$\mathbb{C}[x,t] = \mathbb{C}[P_{\mathrm{in}}, Q_{\mathrm{in}}] \oplus x\mathbb{C}[P_{\mathrm{in}}, Q_{\mathrm{in}}].$$

The sum is direct because the involution $(x,t) \mapsto (-x, t + 2x^p/c)$ fixes $P_{\mathrm{in}}, Q_{\mathrm{in}}$ and sends x to $-x$. Write

$$H = A_{\mathrm{ev}}(P_{\mathrm{in}}, Q_{\mathrm{in}}) + xA_{\mathrm{odd}}(P_{\mathrm{in}}, Q_{\mathrm{in}}).$$

By (6.25), $A_{\mathrm{odd}} \neq 0$. If A_{odd} is not a nonzero multiple of a power of P_{in}, weighted homogeneity makes $A_{\mathrm{odd}}(1,q)$ a nonconstant polynomial in q. Choose a root q_0 and choose y, Y satisfying

$$y^k = (q_0 - 1)/c, \qquad Y^k = (q_0 + 1)/c.$$

The distinct points $(1, y), (-1, Y)$ have equal values of $P_{\text{in}}, Q_{\text{in}}, H$. The collision is isolated. After setting $x = 1$, the first difference equation fixes $X = -1$, the second gives $Y^k - y^k = 2/c$, and the difference of the H values restricts to the nonzero polynomial $2A_{\text{odd}}(1, 1 + cy^k)$. There are only finitely many solutions for y and Y, including when one of them is zero. This contradicts Lemma 3.10. Therefore $A_{\text{odd}} = \kappa_0 P_{\text{in}}^{r_0}$ with $\kappa_0 \neq 0$. Subtracting $A_{\text{ev}}(P, Q)$ from the remainder with initial form H and dividing by κ_0 gives a third coordinate Z with initial form x^N for weights (k, p), where $N = 2r_0 + 1$ is odd. Since $H = y^k\Lambda$ and Λ has positive weight for $\deg x = 1$, $\deg(y^k) = p$, one has $N > p \geq 3$. □

6.6 A collision on a component of the fiber product

Assume that $\boldsymbol{n} = (P, Q, Z)$ satisfies (6.21) and (6.22). We use the weighted initial form of Z to study the difference $Z(x, y) - Z(X, Y)$ on a local component of the fiber product of (P, Q). As in Step 3, injectivity requires this difference to be nonzero at every pair of distinct points. If $\beta(\zeta y) = \beta(y)$ for every $\zeta \in \mu_k$, Weierstrass preparation forces a collision. Otherwise the term of least degree in β whose exponent is not divisible by k determines a second choice of weights, for which injectivity implies a polynomial identity that forces a collision.

Proposition 6.27. *No germ satisfying* (6.21) *and* (6.22) *is injective.*

Proof. Suppose otherwise, and fix a primitive kth root of unity ζ. Write $\beta(y) = \sum_{h\geq 0} \beta_h y^h$, with $\beta_0 = c$.

The local fiber product. Put $\nu = k(p - 2)$ and consider

$$P_\lambda = x^2 + \lambda^\nu y^k, \qquad Q_\lambda = x^p + y^k\beta(\lambda^p y), \qquad R_\lambda = \lambda^{-kN} Z(\lambda^k x, \lambda^p y).$$

These functions are analytic, with $R_0 = x^N$ by (6.22), and for $\lambda \neq 0$ the map $(P_\lambda, Q_\lambda, R_\lambda)$ is obtained from the original germ by invertible scalings. Set $y = 1$ and compare $(s, 1), (S, z)$ near $(0, 1), (0, \zeta)$. The equations for the fiber product of (P_λ, Q_λ) are

$$\begin{aligned} s^2 - S^2 + \lambda^\nu(1 - z^k) &= 0, \\ s^p - S^p + \beta(\lambda^p) - z^k\beta(\lambda^p z) &= 0. \end{aligned} \tag{6.28}$$

The derivative with respect to z of the left-hand side of the second equation at $(s, S, z, \lambda) = (0, 0, \zeta, 0)$ is $-ck\zeta^{k-1} \neq 0$. Thus the second equation has a unique analytic solution $z = \mathfrak{z}(\upsilon, \lambda)$, where $\upsilon = s^p - S^p$. Write

$$1 - \mathfrak{z}(\upsilon, \lambda)^k = d_0(\lambda) + \upsilon\mathcal{K}(\upsilon, \lambda), \qquad K_p = (s^p - S^p)/(s - S).$$

The quotient K_p is a polynomial in s, S. The first equation becomes

$$(s - S)(s + S + \lambda^\nu K_p\mathcal{K}) + \lambda^\nu d_0(\lambda) = 0. \tag{6.29}$$

The functions $U = s - S$, $\Xi = s + S + \lambda^\nu K_p\mathcal{K}$, and λ are analytic coordinates near $(s, S, \lambda) = (0, 0, 0)$, since their Jacobian determinant there is nonzero. On this local fiber product, let $D = R_\lambda(s, 1) - R_\lambda(S, z)$. Since z remains near $\zeta \neq 1$, this neighborhood of the fiber product is disjoint from the diagonal. In particular, $U = 0$ does not describe coincident source points. Injectivity requires the zero set of D to be contained in the set $\lambda = 0$.

If $\beta(\zeta y) = \beta(y)$, uniqueness gives $\mathfrak{z}(0, \lambda) = \zeta$, so $d_0 = 0$. On the component $\Xi = 0$ of the fiber product $U\Xi = 0$, the restriction of D at $\lambda = 0$ is $D|_{\lambda=0} = 2^{1-N}U^N$. Weierstrass preparation [GLS07, Theorem I.1.6] gives zeros of D near $U = 0$ for every small $\lambda \neq 0$. Since z remains near $\zeta \neq 1$, these are collisions of distinct points. We may therefore assume $\beta(\zeta y) \neq \beta(y)$.

Let j be the least index with $\beta_j \neq 0$ and $\zeta^j \neq 1$, and put

$$L = k + j, \qquad \chi = \zeta^j, \qquad T = -\frac{\beta_j}{c} y^L, \qquad \rho = -c(x^2 + T).$$

Comparing coefficients in $\beta(q) = \mathfrak{z}(0,\lambda)^k \beta(q\mathfrak{z}(0,\lambda))$, where $q = \lambda^p$, gives $\mathfrak{z}(0,\lambda) = \zeta + O(q^j)$, since all terms of degree less than j in β are invariant under $y \mapsto \zeta y$. The coefficient of q^j then gives

$$d_0(\lambda) = -\frac{\beta_j}{c}(1-\chi)\lambda^{pj} + O(\lambda^{p(j+1)}). \tag{6.30}$$

Set $r = \nu + pj = pL - 2k$, and write $\lambda^\nu d_0 = \lambda^r \varepsilon(\lambda)$, where $\varepsilon(0) = -(\beta_j/c)(1-\chi) \neq 0$. Replacing Ξ by $-\Xi/\varepsilon(\lambda)$ puts the local fiber product in the form $U\Xi = \lambda^r$. Injectivity excludes zeros of D over $\lambda \neq 0$, while on $\lambda = 0$ its restrictions are $D|_{U=0} = 0$, $D|_{\Xi=0} = 2^{1-N}U^N$.

The two branches $U = 0$ and $\Xi = 0$ of the fiber $\lambda = 0$ correspond respectively to $s = S$ and $s = -S$ when $\lambda = 0$. On the first branch, $z = \mathfrak{z}(0,0) = \zeta$. On the second, $z = \mathfrak{z}(2s^p, 0)$ satisfies $z^k = 1 + 2s^p/c$ and tends to ζ as $s \to 0$. The restrictions of D to the two branches of $\lambda = 0$, together with the absence of zeros when $\lambda \neq 0$, determine its divisor of zeros on $\{U\Xi = \lambda^r\}$. To see this, pull back by the finite cover

$$\pi : (\mathbb{C}^2, 0) \longrightarrow (\{U\Xi = \lambda^r\}, 0), \qquad (\kappa_1, \kappa_2) \longmapsto (\kappa_1^r, \kappa_2^r, \kappa_1\kappa_2).$$

The zeros of $\pi^* D$ lie on the coordinate axes. Unique factorization in $\mathbb{C}\{\kappa_1, \kappa_2\}$ [GLS07, Theorem I.1.16] and the restriction

$$(\pi^* D)|_{\kappa_2 = 0} = 2^{1-N}\kappa_1^{rN}$$

therefore give $\pi^* D = \kappa_1^{rN}\widetilde{\varepsilon}_D$ with $\widetilde{\varepsilon}_D$ a unit. The quotient $\widetilde{\varepsilon}_D = \pi^* D/\kappa_1^{rN}$ is invariant under $(\kappa_1, \kappa_2) \mapsto (\xi\kappa_1, \xi^{-1}\kappa_2)$, $\xi^r = 1$. A monomial $\kappa_1^a \kappa_2^b$ is invariant exactly when r divides $a - b$. Grouping these monomials gives

$$\mathbb{C}\{\kappa_1, \kappa_2\}^{\mu_r} = \mathbb{C}\{\kappa_1^r, \kappa_2^r, \kappa_1\kappa_2\} \cong \mathbb{C}\{U, \Xi, \lambda\}/(U\Xi - \lambda^r).$$

Thus $\widetilde{\varepsilon}_D$ is the pullback $\pi^*\varepsilon_D$ of a unit ε_D on $\{U\Xi = \lambda^r\}$, and

$$D = U^N \varepsilon_D, \qquad \varepsilon_D \in (\mathbb{C}\{U, \Xi, \lambda\}/(U\Xi - \lambda^r))^*. \tag{6.31}$$

Rescaling with weights $(L, 2)$. The substitutions $\lambda = \tau^2$, $s = \tau^r \widehat{s}$, and $S = \tau^r \widehat{S}$ give both $s^2 - S^2$ and $\lambda^r \varepsilon(\lambda)$ in (6.29) a factor τ^{2r}. Since $2k + r = pL$, the points in the original source coordinates become

$$(\tau^{pL}\widehat{s}, \tau^{2p}), \qquad (\tau^{pL}\widehat{S}, \tau^{2p} z),$$

where (x, y) have weights $(L, 2)$ and the scaling parameter is τ^p. After division by τ^{2r}, the term containing $s^p - S^p$ in (6.29) has the factor $\tau^{2\nu + rp - 2r} = \tau^{p(p-2)L}$. Since $p \geq 3$ and $L > 0$, the exponent $p(p-2)L$ is positive, so this term vanishes at $\tau = 0$. Thus (6.30) gives the curve obtained at $\tau = 0$

$$\widehat{S}^2 - \widehat{s}^2 = (1-\chi)T_0, \qquad T_0 = -\beta_j/c \neq 0. \tag{6.32}$$

At each point (u, v) of this curve, $v^2 - u^2 = (1-\chi)T_0 \neq 0$. Thus u, v cannot both vanish, and $u \neq v$. After the substitutions above, the difference of the Q_λ values has nonzero derivative in z at $(\widehat{s}, \widehat{S}, z, \tau) = (u, v, \zeta, 0)$, while the polynomial $\widehat{S}^2 - \widehat{s}^2 - (1-\chi)T_0$ defining the limiting curve (6.32) has gradient $(-2u, 2v)$ in $(\widehat{s}, \widehat{S})$. Fix $\widehat{s} = u$ if $v \neq 0$, or $\widehat{S} = v$ otherwise. The implicit-function theorem gives an analytic lift of (u, v) to the rescaled fiber product, represented by functions

$(\widehat{s}(\tau), \widehat{S}(\tau), z(\tau))$ satisfying the rescaled form of (6.28), with initial value (u, v, ζ). Equation (6.31) implies along every lift

$$\begin{aligned} &Z(\tau^{pL}\widehat{s}, \tau^{2p}) - Z(\tau^{pL}\widehat{S}, \tau^{2p} z) \\ &\qquad = \tau^{pLN}(\widehat{s} - \widehat{S})^N (c_D + O(\tau)), \qquad c_D = \varepsilon_D(0) \neq 0. \end{aligned} \tag{6.33}$$

The coefficient of τ^{pLN} in this difference is $c_D(u - v)^N \neq 0$, so its order is exactly pLN. The constant $c_D = \varepsilon_D(0)$ is independent of the lift.

Let $C_\zeta \subset \mathbb{P}(L, 2, L, 2)$ be the curve defined by $Y = \zeta y$, $\rho(x, y) = \rho(X, Y)$. It lies in $\mathcal{Z}(y^k, \rho)$, and its homogeneous coordinate ring, after eliminating Y, is

$$R = \mathbb{C}[x, y, X]/(X^2 - x^2 - (1 - \chi)T).$$

The ring R is a domain. Indeed, put $\delta_c = X - x$ and $\sigma_c = X + x$. The polynomial defining R becomes $\delta_c\sigma_c - (1 - \chi)T(y)$ in $\mathbb{C}[\delta_c, y][\sigma_c]$. Its coefficients δ_c and $-(1 - \chi)T(y)$ are coprime in $\mathbb{C}[\delta_c, y]$, since $T(y)$ is a nonzero scalar multiple of y^L. Since the polynomial has degree one in σ_c and its coefficients are coprime, Gauss's lemma implies that it is irreducible in $\mathbb{C}[\delta_c, y, \sigma_c]$. Thus the chart $y \neq 0$ of C_ζ is dense. Setting $y = 1$ gives a finite cover of this chart by the curve (6.32).

Put $t = y^k$. The terms of $\beta(y)$ whose exponents are divisible by k sum to $\overline{\beta}(y^k)$, where $\overline{\beta} \in \mathbb{C}\{t\}$ is characterized by $\overline{\beta}(y^k) = k^{-1}\sum_{\eta \in \mu_k} \beta(\eta y)$. For $Q^\flat = Q - P\overline{\beta}(P)$, minimality of j gives

$$\operatorname{in}_{(L,2)} P = t, \qquad \operatorname{in}_{(L,2)} Q^\flat = \rho. \tag{6.34}$$

Indeed, $L = k + j > k$, so y^k has weight $2k < 2L$, the weight of x^2. Since $p \geq 3$, the term x^p has weight $pL > 2L$. The term of least degree in $y^k\beta(y)$ whose exponent is not divisible by k is $\beta_j y^L$, of weight $2L$, and $P\overline{\beta}(P) - t\overline{\beta}(t)$ has initial form cx^2. The map (t, ρ) is finite, and $\mathbb{C}\{P, Q^\flat\} = \mathbb{C}\{P, Q\}$. We may therefore apply Lemma 5.16 with $\mathbb{C}[t, \rho]$ as the algebra generated by the initial forms of $P, Q^\flat$.

For this application, let $H \in \mathbb{C}[x, y]$ and suppose $H(x, y) = H(X, \zeta y)$ in R. The automorphism of R fixing x, y and sending X to $-X$ gives $H(X, \zeta y) = H(-X, \zeta y)$. The inclusion $\mathbb{C}[X, y] \to R$ is injective because the relation defining R can be written as a monic equation in x, so H is even in its first variable. Write $H(x, y) = G(x^2 + T, y)$ with $G \in \mathbb{C}[\mathcal{S}, y]$. Since $X^2 + \chi T = x^2 + T$, the identity $H(x, y) = H(X, \zeta y)$ becomes

$$G(x^2 + T, y) = G(x^2 + T, \zeta y).$$

The homomorphism

$$\mathbb{C}[\mathcal{S}, y] \longrightarrow \mathbb{C}[x, y], \qquad \mathcal{S} \longmapsto x^2 + T, \quad y \longmapsto y$$

is injective. Since ζ is a primitive kth root of unity, $G \in \mathbb{C}[\mathcal{S}, y^k]$, and hence $H \in \mathbb{C}[\rho, t]$. We have proved

$$H(x, y) = H(X, \zeta y) \text{ in } R \quad \Longrightarrow \quad H \in \mathbb{C}[t, \rho]. \tag{6.35}$$

By (6.24), the map (P, Q) has degree $2k > 1$. Injectivity therefore excludes $Z \in \mathbb{C}\{P, Q\}$. Thus Lemma 5.16, applied to Z, gives a remainder $\widetilde{Z}$ with $H = \operatorname{in}_{(L,2)} \widetilde{Z} \notin \mathbb{C}[t, \rho]$. Write $\delta = \operatorname{ord}_{(L,2)} \widetilde{Z}$. By (6.35), the difference $H(x, y) - H(X, \zeta y)$ is nonzero at some $(u, 1), (v, \zeta)$ on the dense chart of C_ζ. On the analytic lift of this pair the difference of the $\widetilde{Z}$ values has order $p\delta$ in τ. Subtracting functions of P, Q preserves the difference of the Z values on the fiber product of (P, Q), so (6.33) gives $\delta = NL$. We compare the coefficients of τ^{pLN} along the analytic lift of every point of (6.32). The polynomials $H(x, y) - H(X, \zeta y)$ and $c_D(x - X)^N$ both have weight NL for weights $(L, 2, L, 2)$, so equality for $y = 1$ extends to the dense chart $y \neq 0$ and hence to all of C_ζ:

$$H(x, y) - H(X, \zeta y) = c_D(x - X)^N \quad \text{on } C_\zeta. \tag{6.36}$$

The final collision. Only the odd part of H in x can distinguish (x, y) from $(-x, y)$. Since H has weight NL and N is odd by (6.22), every odd power x^i in H has y-exponent $L(N-i)/2$, a multiple of L. Thus its odd part can be written

$$\frac{H(x,y) - H(-x,y)}{2} = A_N(x,T), \qquad \deg x = 1, \quad \deg T = 2, \quad \deg A_N = N.$$

The simultaneous sign change $(x, X) \mapsto (-x, -X)$ preserves C_ζ and negates the right-hand side of (6.36). Subtracting the transformed identity from the original gives

$$A_N(x,T) - A_N(X, \chi T) = c_D(x-X)^N \quad \text{when } X^2 - x^2 = (1-\chi)T. \tag{6.37}$$

Setting $X = -x$ and $T = 0$ gives

$$A_N(x,0) = 2^{N-1} c_D x^N.$$

The order N of $c_D(x-X)^N$ along $X = x$ also constrains the partial derivatives of A_N at $T = 0$. Substitute

$$X = x + h, \qquad T = \frac{2xh + h^2}{1-\chi}$$

and differentiate (6.37) at $h = 0$. Since $N \geq 3$, the derivative of its right-hand side is zero, so

$$2x\,\partial_T A_N(x,0) - \partial_x A_N(x,0) = 0, \qquad \partial_T A_N(x,0) = 2^{N-2} N c_D x^{N-2}.$$

For $T_0 = -\beta_j/c \neq 0$, the odd polynomial $A_N(x, T_0)$ therefore has nonzero terms of degrees N and $N-2$. Dividing by its lowest power of x leaves a polynomial of positive degree with nonzero constant term, hence a nonzero root x_0. The initial forms in (6.34) take the same values at $(x_0, 1), (-x_0, 1)$, and the difference of the H values is $2A_N(x_0, T_0) = 0$. After setting $y = 1$, the collision is isolated, since the first difference equation for (t, ρ, H) fixes $Y = 1$, the second fixes $X = -x$ near $x_0 \neq 0$, and the difference of the H values restricts to the nonzero polynomial $2A_N(x, T_0)$. This contradicts Lemma 3.10. □

Theorem 6.38. *Under the hypotheses of Section 2, one has $m_{\boldsymbol{n}} \neq 2$.*

Proof. Proposition 6.18 reduces the case $m_{\boldsymbol{n}} = 2$ to (6.20). Proposition 6.23 gives (6.21) and (6.22), which Proposition 6.27 excludes. □

Proof of Theorem 1.1. If $(d\boldsymbol{n})_0 = 0$, the reduction theorem [PC26, Theorem 1.1] gives $m_{\boldsymbol{n}} \in \{2, 3, 4\}$. Theorem 4.1 excludes four. Theorems 5.17 and 6.38 exclude three and two, respectively. □

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